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MASTER'S THESIS

submitted in partial fulfilment of the requirements for the degree of

**Master of Engineering (M.Eng.)**

Advanced Industrial Engineering

Technische Hochschule Rosenheim

# **Achieving Data Sovereignty for EU Enterprises Through European Cloud Infrastructure**

A Comparative Analysis of Technical Capabilities  
and Regulatory Compliance

in cooperation with

*Alps Alpine Europe GmbH*

*Local Products Department*

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Munich, September 21, 2026

# Abstract

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# Dedication

thank you everyone for the support Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

# Acknowledgment

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# Contents

|                                                                                    |           |
|------------------------------------------------------------------------------------|-----------|
| <b>List of Abbreviations</b>                                                       | <b>11</b> |
| <b>1 Introduction</b>                                                              | <b>13</b> |
| 1.1 Background and Motivation . . . . .                                            | 13        |
| 1.2 Problem Statement . . . . .                                                    | 13        |
| 1.3 Research Questions . . . . .                                                   | 13        |
| 1.4 Scope and Limitations . . . . .                                                | 13        |
| 1.5 Industry Partnership Context (Alps Alpine Europe GmbH) . . . . .               | 13        |
| 1.6 Thesis Structure . . . . .                                                     | 13        |
| <b>2 Fundamentals</b>                                                              | <b>14</b> |
| 2.1 Cloud Computing Paradigms (IaaS, PaaS, SaaS) . . . . .                         | 14        |
| 2.2 Data Sovereignty vs. Data Residency — Key Distinctions . . . . .               | 14        |
| 2.3 Cloud Architectural Philosophies . . . . .                                     | 14        |
| 2.3.1 Centralised (AWS, GCP, Azure) . . . . .                                      | 14        |
| 2.3.2 Decentralised (P2P, Blockchain-based) . . . . .                              | 14        |
| 2.3.3 Federated (GAIA-X, Catena-X) . . . . .                                       | 14        |
| 2.3.4 Self-Hosted Infrastructure . . . . .                                         | 14        |
| 2.4 Vendor Lock-in and Strategic Dependency . . . . .                              | 14        |
| 2.5 EU Regulatory Landscape . . . . .                                              | 14        |
| 2.5.1 GDPR . . . . .                                                               | 15        |
| 2.5.2 Schrems II (Case C-311/18) . . . . .                                         | 15        |
| 2.5.3 CLOUD Act . . . . .                                                          | 15        |
| 2.5.4 FISA Section 702 and RISAA 2024 . . . . .                                    | 15        |
| 2.5.5 EU Data Act (Regulation 2023/2854) . . . . .                                 | 15        |
| 2.6 The EU Digital Decade and the 2030 Digital Compass . . . . .                   | 15        |
| 2.7 The Jurisdictional Problem — Why Data Location Alone Is Insufficient . . . . . | 15        |
| <b>3 State of the Art</b>                                                          | <b>16</b> |
| 3.1 Prior Work on Cloud Provider Benchmarking . . . . .                            | 16        |
| 3.2 Existing Data Sovereignty Frameworks and Assessments . . . . .                 | 16        |
| 3.3 GAIA-X and Its Limitations . . . . .                                           | 16        |
| 3.4 European Cloud Market Landscape . . . . .                                      | 16        |
| 3.5 Gap Analysis — What This Thesis Addresses . . . . .                            | 16        |

|          |                                                                                  |           |
|----------|----------------------------------------------------------------------------------|-----------|
| <b>4</b> | <b>Provider Selection and Compliance Analysis</b>                                | <b>17</b> |
| 4.1      | Selection Criteria . . . . .                                                     | 17        |
| 4.2      | Provider Profiles by Country . . . . .                                           | 18        |
| 4.2.1    | Germany . . . . .                                                                | 18        |
| 4.2.2    | France . . . . .                                                                 | 20        |
| 4.2.3    | Sweden . . . . .                                                                 | 21        |
| 4.2.4    | Switzerland . . . . .                                                            | 21        |
| 4.2.5    | The Netherlands . . . . .                                                        | 22        |
| 4.2.6    | Finland . . . . .                                                                | 22        |
| 4.2.7    | On-Premises Reference . . . . .                                                  | 23        |
| 4.3      | Intra-Provider Comparison: OTC and T Cloud Public . . . . .                      | 23        |
| 4.4      | AWS as Reference Baseline . . . . .                                              | 25        |
| 4.5      | Summary . . . . .                                                                | 25        |
| <b>5</b> | <b>Methodology</b>                                                               | <b>27</b> |
| 5.1      | Research Design and Overall Approach . . . . .                                   | 27        |
| 5.2      | CloudSov Tool: Implementation and Usage . . . . .                                | 27        |
| 5.2.1    | Overview and Availability . . . . .                                              | 27        |
| 5.2.2    | Getting Started . . . . .                                                        | 28        |
| 5.2.3    | Interface Walkthrough . . . . .                                                  | 28        |
| 5.2.4    | Advantages, Limitations, and Controversies . . . . .                             | 29        |
| 5.3      | Sovereignty Scoring Framework (SovScore) . . . . .                               | 31        |
| 5.3.1    | Theoretical Basis and Framework Selection . . . . .                              | 31        |
| 5.3.2    | SOV Objectives and Weighting Scheme . . . . .                                    | 31        |
| 5.3.3    | Evidence Collection: The Y/P/N Verdict Protocol . . . . .                        | 33        |
| 5.3.4    | SEAL Level Computation . . . . .                                                 | 34        |
| 5.3.5    | Minimum SEAL Thresholds and Pass/Fail Determination . . . . .                    | 34        |
| 5.3.6    | Provider Coverage and Tool Implementation . . . . .                              | 35        |
| 5.3.7    | Preliminary SovScore Findings . . . . .                                          | 36        |
| 5.4      | Cloud Readiness Assessment Design . . . . .                                      | 36        |
| 5.4.1    | Rationale and Baseline Selection . . . . .                                       | 36        |
| 5.4.2    | Service Taxonomy: 11 Categories and 60 Services . . . . .                        | 37        |
| 5.4.3    | Coverage Scoring Protocol . . . . .                                              | 37        |
| 5.4.4    | Dimension Score Computation . . . . .                                            | 38        |
| 5.4.5    | Tool Implementation and Visualisation . . . . .                                  | 38        |
| 5.4.6    | Preliminary Readiness Findings . . . . .                                         | 39        |
| 5.5      | Financial Analysis Methodology . . . . .                                         | 40        |
| 5.5.1    | Rationale and Scope . . . . .                                                    | 40        |
| 5.5.2    | Data Sources and Collection . . . . .                                            | 40        |
| 5.5.3    | Investment Milestone Classification . . . . .                                    | 41        |
| 5.5.4    | Financial Consideration Module: Tool Implementation and Visualisations . . . . . | 41        |
| 5.5.5    | Preliminary Financial Findings . . . . .                                         | 44        |
| 5.5.6    | Limitations . . . . .                                                            | 45        |
| 5.6      | Benchmark Pipeline Design . . . . .                                              | 45        |
| 5.6.1    | Compute (sysbench) . . . . .                                                     | 46        |
| 5.6.2    | Storage (fio) . . . . .                                                          | 46        |

|          |                                                                                               |           |
|----------|-----------------------------------------------------------------------------------------------|-----------|
| 5.6.3    | Network (iperf3)                                                                              | 46        |
| 5.6.4    | AI/ML Inference (Odin)                                                                        | 46        |
| 5.7      | Cost Evaluation Methodology (TCO)                                                             | 46        |
| 5.8      | Deployment Experiment Design (Phase 3)                                                        | 46        |
| 5.9      | Evaluation Criteria and Metrics Summary                                                       | 46        |
| <b>6</b> | <b>Technical Benchmarking and Price Consideration</b>                                         | <b>47</b> |
| <b>7</b> | <b>Deployment Experiment</b>                                                                  | <b>50</b> |
| 7.1      | Selected Provider and Rationale                                                               | 50        |
| 7.2      | EU Stack Setup (Mosquitto/EMQX, FastAPI, Grafana, Prometheus)                                 | 50        |
| 7.3      | AWS Stack Setup (IoT Core, SageMaker, CloudWatch)                                             | 50        |
| 7.4      | Workload Simulation and Results                                                               | 50        |
| 7.5      | Latency Analysis (p50, p95, p99)                                                              | 50        |
| 7.6      | Operational Friction Log                                                                      | 50        |
| 7.7      | Cost Comparison                                                                               | 51        |
| <b>8</b> | <b>Discussion</b>                                                                             | <b>52</b> |
| 8.1      | Research Question Responses                                                                   | 52        |
| 8.1.1    | RQ1: Can EU cloud providers satisfy the EC DG DIGIT Cloud Sovereignty Framework requirements? | 52        |
| 8.1.2    | RQ2: How does the readiness of EU providers compare to AWS?                                   | 53        |
| 8.1.3    | RQ3: What is the trade-off between sovereignty and readiness?                                 | 54        |
| 8.2      | The Sovereignty–Readiness Trade-off                                                           | 55        |
| 8.3      | The Huawei Problem and Deutsche Telekom’s Strategic Response                                  | 57        |
| 8.3.1    | The Legal Exposure Under SOV-2                                                                | 57        |
| 8.3.2    | The Hardware and Firmware Exposure Under SOV-5                                                | 58        |
| 8.3.3    | Deutsche Telekom’s Strategic Response                                                         | 58        |
| 8.3.4    | What the SovScore Assessment Captured                                                         | 58        |
| 8.3.5    | The OTC–T Cloud Public Intra-Provider Comparison                                              | 59        |
| 8.4      | Timeline and Investment: AWS versus EU Providers                                              | 60        |
| 8.4.1    | The Cloud Maturity Trajectory                                                                 | 60        |
| 8.4.2    | IPCEI-CIS and the European Cloud Strategy                                                     | 60        |
| 8.4.3    | The AI/ML Horizon                                                                             | 61        |
| 8.5      | Financial Landscape and Capital Efficiency of EU Sovereign Cloud Providers                    | 61        |
| 8.5.1    | The Revenue Chasm: Scale Asymmetry Between AWS and EU Providers                               | 61        |
| 8.5.2    | Readiness Gap Closure: Projected Parity Timelines                                             | 63        |
| 8.5.3    | Investment Milestones and Capital Events                                                      | 64        |
| 8.5.4    | Capital Efficiency: Readiness Points per Euro Invested                                        | 65        |
| 8.5.5    | Capital Structure and its Sovereignty Implications                                            | 66        |
| 8.5.6    | Capital Source as a Direct Sovereignty Determinant                                            | 67        |
| 8.6      | Performance vs. Sovereignty: Structural Constraints                                           | 68        |
| 8.6.1    | SOV-6 Q4: No EU HPC Silicon                                                                   | 68        |
| 8.6.2    | SOV-5 Q2: Non-EU Firmware                                                                     | 68        |
| 8.6.3    | Policy Instruments                                                                            | 69        |
| 8.7      | Vendor Lock-in Implications                                                                   | 69        |
| 8.7.1    | OpenStack as an Interoperability Anchor                                                       | 69        |

|          |                                                                           |           |
|----------|---------------------------------------------------------------------------|-----------|
| 8.7.2    | Proprietary Managed Services and Lock-in . . . . .                        | 70        |
| 8.7.3    | The Migration Barrier from AWS . . . . .                                  | 70        |
| 8.8      | Recommendations for EU Enterprises . . . . .                              | 70        |
| 8.8.1    | Strategy 1: Lift-and-Shift to a High-Readiness EU Provider . . . . .      | 70        |
| 8.8.2    | Strategy 2: Sovereignty-First Selection for Sensitive Workloads . . . . . | 71        |
| 8.8.3    | Strategy 3: Hybrid Cloud Architecture . . . . .                           | 71        |
| 8.8.4    | Strategy 4: Monitor STACKIT for 2027–2028 Re-evaluation . . . . .         | 71        |
| 8.9      | Limitations of This Study . . . . .                                       | 71        |
| 8.9.1    | Self-Reported and Publicly Documented Evidence Base . . . . .             | 71        |
| 8.9.2    | Static Snapshot in a Rapidly Evolving Market . . . . .                    | 72        |
| 8.9.3    | AWS Baseline Bias . . . . .                                               | 72        |
| 8.9.4    | Service Selection Tailored to a Specific Enterprise . . . . .             | 72        |
| 8.9.5    | No SLA or Operational Quality Dimension . . . . .                         | 72        |
| 8.9.6    | Exclusion of Cost and Commercial Terms . . . . .                          | 73        |
| <b>9</b> | <b>Conclusion</b>                                                         | <b>74</b> |
| 9.1      | Summary of Findings . . . . .                                             | 74        |
| 9.2      | Contribution to Research . . . . .                                        | 74        |
| 9.3      | Future Work . . . . .                                                     | 74        |
| <b>A</b> | <b>Alps Alpine Infrastructure Checklist (Full)</b>                        | <b>75</b> |
| <b>B</b> | <b>Europeanness Scoring Matrix</b>                                        | <b>76</b> |
| <b>C</b> | <b>Benchmark Raw Data</b>                                                 | <b>77</b> |
| <b>D</b> | <b>Terraform Provisioning Scripts</b>                                     | <b>78</b> |
| <b>E</b> | <b>Operational Friction Log (Full)</b>                                    | <b>79</b> |

# List of Figures

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                   |    |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 5.1  | CloudSov SovScore module: provider grid with verdict inputs and computed scores.                                                                                                                                                                                                                                                                                                                                                  | 29 |
| 5.2  | CloudSov Readiness Assessment module: dimension scores across providers. . . . .                                                                                                                                                                                                                                                                                                                                                  | 30 |
| 5.3  | CloudSov gap table: services ranked by average deficit from the AWS baseline. . . .                                                                                                                                                                                                                                                                                                                                               | 30 |
| 5.4  | SovScore web interface: provider assessment grid with SEAL levels and overall sovereignty scores. . . . .                                                                                                                                                                                                                                                                                                                         | 35 |
| 5.5  | Min-SEAL justification tooltip on the SovScore grid header. . . . .                                                                                                                                                                                                                                                                                                                                                               | 35 |
| 5.6  | Cloud Readiness Assessment: coverage profile radar (left) and dimension breakdown area chart (right). . . . .                                                                                                                                                                                                                                                                                                                     | 39 |
| 5.7  | Gap summary table with an expanded provider row showing per-service EU alternatives. . . . .                                                                                                                                                                                                                                                                                                                                      | 39 |
| 5.8  | Readiness gap closure: historical readiness velocities (solid) projected forward (dashed) to AWS parity (score = 100). The amber vertical line marks the EC Digital Decade 2030 target [11]. . . . .                                                                                                                                                                                                                              | 42 |
| 5.9  | Annual cloud-segment revenue (EUR M, log scale): AWS vs the five EU providers, 2019–2024. A log scale is required to render both AWS and EU providers on a common axis. . . . .                                                                                                                                                                                                                                                   | 43 |
| 5.10 | Investment and funding milestones for AWS and the five EU providers, 2006–2026. Bubble area is proportional to the committed capital (EUR); dashed outlines indicate undisclosed amounts. . . . .                                                                                                                                                                                                                                 | 43 |
| 5.11 | Investment efficiency scatter: current average readiness score vs cumulative disclosed capital (log scale, €M). The efficiency ratio $\eta$ (readiness points per €B) is shown beneath each provider label. . . . .                                                                                                                                                                                                               | 44 |
| 5.12 | Capital structure by funding source: total disclosed capital decomposed by category for each EU provider (€M). Undisclosed events are excluded. . . . .                                                                                                                                                                                                                                                                           | 45 |
| 8.1  | Sovereignty–readiness plane for the five EU providers under financial analysis, plotted using actual computed SovScore and average cloud readiness scores. AWS (readiness baseline = 100%) and a locally hosted boundary marker are shown for reference. The dashed line marks the SEAL 4 SovScore threshold (80%). No provider currently occupies the upper-right quadrant (high sovereignty <i>and</i> high readiness). . . . . | 56 |
| 8.2  | Annual cloud-segment revenue (EUR M, log scale) for AWS and the five EU providers, 2019–2024. A log scale is required to render both AWS and the EU providers on a common axis; on a linear scale EU provider lines would be indistinguishable from zero. STACKIT data begins at 2022 (launch 2021, revenue not publicly disclosed in first year). . . . .                                                                        | 62 |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                     |    |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 8.3 | Historical readiness trajectories (solid lines) modelled as linear growth from zero at each provider’s cloud launch year to the Q2 2026 average readiness score, with projections (dashed lines) at the same velocity to AWS parity (score = 100). Vertical markers indicate estimated parity years. . . . .                                                                                                        | 63 |
| 8.4 | Investment and funding milestones for AWS and the five EU providers, 2012–2026. Bubble area is proportional to the committed amount (EUR); colour encodes the provider. AWS milestones are included for scale reference. . . . .                                                                                                                                                                                    | 64 |
| 8.5 | Investment efficiency scatter: each provider is plotted by cumulative disclosed capital (EUR M, log scale) on the X-axis and current average readiness score on the Y-axis. The efficiency ratio (readiness points per €B of disclosed capital) is shown beneath each label. STACKIT’s low ratio reflects that its €11B Bitterfeld-Wolfen commitment is a forward pledge, not yet fully deployed. . . . .           | 65 |
| 8.6 | Capital structure decomposition for the five EU providers, showing cumulative disclosed capital by funding source category (EUR M). T-Systems abbreviates T-Systems (OTC / T-Cloud Public). STACKIT’s bar is dominated by the €11B Schwarz Group parent pledge at Bitterfeld-Wolfen. OVHcloud and IONOS are the only two providers with US-managed-fund exposure in the Market Capital and Debt categories. . . . . | 66 |

# List of Tables

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |    |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 4.1 | Structural comparison of T-Systems OTC and T Cloud Public . . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 23 |
| 4.2 | Summary of selected EU cloud providers . . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 26 |
| 5.1 | Interface annotation legend — numbered regions shown in Figures 5.1–5.3 . . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 28 |
| 5.2 | SOV objectives, weights, question counts, and minimum SEAL thresholds . . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 32 |
| 5.3 | SOV objectives SOV-1 to SOV-4: weights, min. SEAL thresholds, and evaluation questions . . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 32 |
| 5.4 | SOV objectives SOV-5 to SOV-8: weights, min. SEAL thresholds, and evaluation questions . . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 33 |
| 5.5 | SEAL level thresholds and descriptions . . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 34 |
| 5.6 | All 60 AWS services assessed across 13 readiness dimensions ( <sup>†</sup> = industry-standard; not specifically required by Alps Alpine) . . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 37 |
| 8.1 | SovScore results for the five EU providers under financial analysis: per-objective SEAL levels (0–4), weighted aggregate score (0–100), and pass/fail determination against minimum SEAL thresholds. Superscript <sup>†</sup> denotes a cell below the minimum threshold for that objective. Minimum SEAL thresholds per objective: SOV-1 $\geq$ 3, SOV-2 $\geq$ 4, SOV-3 $\geq$ 3, SOV-4 $\geq$ 2, SOV-5 $\geq$ 2, SOV-6 $\geq$ 2, SOV-7 $\geq$ 3, SOV-8 $\geq$ 1. A provider passes only if all eight thresholds are met simultaneously. T-Systems = T-Systems (OTC / T-Cloud Public), merged readiness entity. . . . . | 53 |
| 8.2 | Cloud Readiness Assessment dimension scores (0–100) for the five EU providers. Providers are ordered by average readiness score (highest to lowest). Scalability and Performance are pre-scored meta-dimensions; all other dimensions are computed from service-level coverage verdicts. T-Systems = T-Systems (OTC / T-Cloud Public) merged entity, using T-Cloud Public assessment data. . . . .                                                                                                                                                                                                                        | 54 |

# List of Abbreviations

|                  |                                                          |
|------------------|----------------------------------------------------------|
| <b>IaaS</b>      | Infrastructure as a Service                              |
| <b>PaaS</b>      | Platform as a Service                                    |
| <b>SaaS</b>      | Software as a Service                                    |
| <b>API</b>       | Application Programming Interface                        |
| <b>SLA</b>       | Service Level Agreement                                  |
| <b>TCO</b>       | Total Cost of Ownership                                  |
| <b>PUE</b>       | Power Usage Effectiveness                                |
| <b>HPC</b>       | High-Performance Computing                               |
| <b>SOV</b>       | Sovereignty Objective                                    |
| <b>SEAL</b>      | Sovereignty Assurance Level                              |
| <b>EUCS</b>      | EU Cybersecurity Certification Scheme for Cloud Services |
| <b>CMEK</b>      | Customer-Managed Encryption Keys                         |
| <b>KMS</b>       | Key Management Service                                   |
| <b>IAM</b>       | Identity and Access Management                           |
| <b>GDPR</b>      | General Data Protection Regulation                       |
| <b>NIS2</b>      | Network and Information Security Directive (Revision 2)  |
| <b>DORA</b>      | Digital Operational Resilience Act                       |
| <b>CSRD</b>      | Corporate Sustainability Reporting Directive             |
| <b>CLOUD Act</b> | Clarifying Lawful Overseas Use of Data Act               |
| <b>EU</b>        | European Union                                           |
| <b>EC</b>        | European Commission                                      |
| <b>DIGIT</b>     | Directorate-General for Digital Services                 |

|               |                                                     |
|---------------|-----------------------------------------------------|
| <b>ENISA</b>  | European Union Agency for Cybersecurity             |
| <b>CJEU</b>   | Court of Justice of the European Union              |
| <b>BSI</b>    | Bundesamt für Sicherheit in der Informationstechnik |
| <b>NIST</b>   | National Institute of Standards and Technology      |
| <b>ISO</b>    | International Organization for Standardization      |
| <b>AWS</b>    | Amazon Web Services                                 |
| <b>GCP</b>    | Google Cloud Platform                               |
| <b>GAIA-X</b> | European Cloud and Data Infrastructure Initiative   |
| <b>SOC</b>    | Service Organisation Control                        |
| <b>IP</b>     | Intellectual Property                               |
| <b>OTC</b>    | Open Telekom Cloud                                  |
| <b>TISAX</b>  | Trusted Information Security Assessment Exchange    |
| <b>C5</b>     | Cloud Computing Compliance Criteria Catalogue (BSI) |
| <b>IPCEI</b>  | Important Project of Common European Interest       |
| <b>EIB</b>    | European Investment Bank                            |
| <b>IPO</b>    | Initial Public Offering                             |
| <b>CAGR</b>   | Compound Annual Growth Rate                         |
| <b>PE</b>     | Private Equity                                      |
| <b>VC</b>     | Venture Capital                                     |
| <b>DAX</b>    | Deutscher Aktienindex                               |
| <b>CAPEX</b>  | Capital Expenditure                                 |

# Chapter 1

## Introduction

### 1.1 Background and Motivation

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### 1.2 Problem Statement

.....i can reference a quote here..... [22]

I want to do a footnote right here <sup>1</sup>

### 1.3 Research Questions

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### 1.4 Scope and Limitations

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### 1.5 Industry Partnership Context (Alps Alpine Europe GmbH)

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### 1.6 Thesis Structure

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<sup>1</sup>this is my firstnote!

## Chapter 2

# Fundamentals

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### 2.1 Cloud Computing Paradigms (IaaS, PaaS, SaaS)

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### 2.2 Data Sovereignty vs. Data Residency — Key Distinctions

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### 2.3 Cloud Architectural Philosophies

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#### 2.3.1 Centralised (AWS, GCP, Azure)

#### 2.3.2 Decentralised (P2P, Blockchain-based)

#### 2.3.3 Federated (GAIA-X, Catena-X)

#### 2.3.4 Self-Hosted Infrastructure

### 2.4 Vendor Lock-in and Strategic Dependency

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### 2.5 EU Regulatory Landscape

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### 2.5.1 GDPR

### 2.5.2 Schrems II (Case C-311/18)

### 2.5.3 CLOUD Act

### 2.5.4 FISA Section 702 and RISAA 2024

### 2.5.5 EU Data Act (Regulation 2023/2854)

## 2.6 The EU Digital Decade and the 2030 Digital Compass

The EU Digital Decade Policy Programme [11], established by Decision (EU) 2022/2481 of the European Parliament and of the Council, sets the Union’s overarching digital transformation agenda through 2030. Structured around four cardinal points — digital skills, digital infrastructure, digitalisation of businesses, and digitalisation of public services — the programme translates the Commission’s 2021 Digital Compass communication into legally binding trajectory targets and an annual monitoring mechanism. For cloud computing specifically, the programme targets 75% of EU enterprises using cloud services, artificial intelligence, or big data by 2030, and at least 80% of the EU population holding basic digital skills [11]. A parallel target requires EU Member States to migrate all digitisable public services online and to offer them interoperably across borders by the same year.

The relevance of the Digital Decade to cloud sovereignty is direct. If 75% of EU enterprises and the majority of Member State administrations are to operate on cloud infrastructure by 2030, the origin, jurisdiction, and governance of that infrastructure become policy questions of the first order. An EU enterprise that meets the Digital Decade cloud target by migrating workloads to a US-headquartered hyperscaler may satisfy the letter of the programme’s digitalisation objective while simultaneously creating the CLOUD Act exposure and data-transfer vulnerabilities discussed in Section ???. This tension — between *digital adoption* as measured by the Digital Decade scoreboard and *digital sovereignty* as assessed by frameworks such as the one developed in this thesis — is the principal motivation for examining EU-origin cloud providers not merely as a market segment, but as a strategic infrastructure class.

## 2.7 The Jurisdictional Problem — Why Data Location Alone Is Insufficient

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## Chapter 3

# State of the Art

### 3.1 Prior Work on Cloud Provider Benchmarking

...

### 3.2 Existing Data Sovereignty Frameworks and Assessments

...

### 3.3 GAIA-X and Its Limitations

...

### 3.4 European Cloud Market Landscape

...

### 3.5 Gap Analysis — What This Thesis Addresses

...

## Chapter 4

# Provider Selection and Compliance Analysis

### 4.1 Selection Criteria

The 18 European cloud providers studied in this thesis were selected according to a set of criteria designed to ensure that each provider represents a genuinely sovereign European cloud option that can be assessed consistently against the EC DG DIGIT Cloud Sovereignty Framework v1.2.1 [13] and the 60-service Cloud Readiness Assessment (see Chapter 5). The criteria are as follows.

1. **European legal domicile.** The provider must be headquartered and legally registered in a European Union or European Economic Area (EEA) member state. This criterion reflects the jurisdictional foundation of digital sovereignty: legal recourse, data-protection obligations, and government-access demands are all governed by the law of the country in which the provider is incorporated.
2. **Commercial cloud service offering.** The provider must offer publicly available commercial cloud services at a minimum of the Infrastructure-as-a-Service (IaaS) or Platform-as-a-Service (PaaS) tier. Providers offering only managed hosting, colocation, or pure Software-as-a-Service products without an underlying programmable infrastructure layer were excluded, as the readiness assessment is structured around service-level parity with AWS.
3. **Documented service catalogue.** The provider must maintain a publicly accessible service catalogue sufficient to determine which of the 60 assessed service categories are available. Providers for which no structured catalogue could be identified in English or in a machine-translatable form were excluded to preserve assessment consistency.
4. **European data centre operations.** All data processing and storage must be operated from data centres physically located in Europe. Providers that route or replicate data to non-European facilities as a default were excluded.
5. **Operational independence from US hyperscalers.** The provider must be independently operated and must not function as a white-label resale or thin wrapper around an AWS, Azure,

or Google Cloud back-end. Resellers and managed-service partners of US hyperscalers were excluded because their sovereignty profile is entirely dependent on the underlying hyperscaler’s terms, and assessing them separately would be redundant.

6. **Sufficient public documentation.** The provider must have published enough technical and governance documentation to allow the EC DG DIGIT framework questions to be answered reliably from open sources. This criterion operationalises transparency as a precondition for sovereignty: a provider that does not disclose its data-handling practices cannot credibly claim a high sovereignty posture.

Amazon Web Services (AWS) is included separately as the universal readiness baseline, not as a selected EU provider. Its role in the assessment framework is described in Section 4.4.

## 4.2 Provider Profiles by Country

The 18 selected providers span six countries. The following subsections group them geographically and provide a brief profile of each provider’s founding, ownership, data centre geography, primary service tier, and rationale for inclusion.

### 4.2.1 Germany

#### **Hetzner**

Founded in 1997 and headquartered in Gunzenhausen, Bavaria, Hetzner Online GmbH is a family-owned German company that has grown into one of Europe’s largest independent data centre operators [19]. Its primary data centre campuses are located in Nuremberg, Falkenstein, and Helsinki, with additional infrastructure in Ashburn (US) for non-EU markets. Hetzner’s core offering is bare-metal dedicated servers and cloud virtual machines (IaaS), complemented by storage boxes and load balancers. Its European data residency is guaranteed by contract, and its independence from any US parent company eliminates the CLOUD Act exposure that affects hyperscaler subsidiaries. Hetzner was selected for this study because of its large and growing European user base, its highly competitive pricing, and its status as a reference point for cost-conscious EU sovereignty.

#### **IONOS**

Originally founded in 1988 as 1&1 Internet, IONOS was rebranded as a standalone cloud brand in 2018 [21]. It is owned by United Internet AG, a Frankfurt Stock Exchange-listed German holding company, and is headquartered in Montabaur, Germany. IONOS operates data centres across Germany, Spain, the United Kingdom, and the United States, with its European offering centred on IaaS compute, managed virtual private servers, object storage, and a range of website and e-commerce tools. As one of the largest European hosting and cloud incumbents by revenue and customer count, IONOS was selected to represent the established German IaaS market and to test whether scale translates into a broader managed-service catalogue comparable to US hyperscalers.

#### **STACKIT**

STACKIT was launched in 2021 as the enterprise cloud brand of Schwarz IT, the technology arm of the Schwarz Group — the parent company of the Lidl and Kaufland retail chains and one of the

largest private companies in Europe [30]. STACKIT is headquartered in Heilbronn, Germany, and operates its infrastructure exclusively from German and European data centres. Its offering targets enterprise and regulated-sector customers with IaaS compute, managed Kubernetes, object storage, and data analytics services, all backed by the financial resources of the Schwarz Group. STACKIT was selected as a notable new entrant that brings significant European industrial capital into the cloud market and represents the emerging category of retail-conglomerate-owned sovereign clouds.

### **T-Systems Open Telekom Cloud**

The Open Telekom Cloud (OTC) is the public cloud offering of T-Systems International GmbH, the enterprise ICT subsidiary of Deutsche Telekom AG [8]. Launched approximately in 2016 and operated from data centres in Biere and Magdeburg (Saxony-Anhalt, Germany), OTC is built on Huawei FusionSphere — Huawei’s proprietary OpenStack distribution — and offers a broad IaaS and PaaS catalogue including compute, storage, networking, database services, and container management. It holds a BSI C5 attestation, making it one of the most compliance-certified European cloud offerings available, and is widely used for German federal and state government workloads. OTC was selected for the depth of its compliance documentation and its relevance to public-sector sovereignty procurement decisions.

### **Arvato Systems**

Arvato Systems is the information technology subsidiary of Bertelsmann SE & Co. KGaA, the Gütersloh-based German media and services conglomerate [4]. Rather than operating a self-branded public cloud infrastructure platform, Arvato Systems delivers managed cloud, hybrid infrastructure, and B2B IT services primarily to enterprise clients in media, healthcare, and logistics. Its European data centres support GDPR-compliant processing under German and EU law. Arvato Systems was selected as a representative of large enterprise managed-cloud providers that are deeply embedded in German industry but do not have a fully self-built hyperscaler-equivalent infrastructure, thereby testing how managed-service specialists score in the readiness assessment.

### **Noris Network**

Founded in 1993 and headquartered in Nuremberg, Germany, Noris Network AG specialises in high-security data centre infrastructure and managed cloud services for regulated industries [26]. The company operates its own data centres in Nuremberg, certified to ISO 27001, and offers managed hosting, private cloud, and hybrid connectivity solutions. Its customer base is concentrated in finance, healthcare, and critical infrastructure sectors, where compliance and physical control of infrastructure are paramount. Noris Network was selected as a mid-size German provider whose strong compliance posture and sector-specific expertise represent an important segment of the European sovereign cloud market that sits between hyperscalers and commodity hosting.

### **PlusServer**

PlusServer GmbH is a Cologne-based cloud and hosting provider that forms part of the Ioniq Group [28]. Its service portfolio covers sovereign cloud (built on VMware and Sovereign Cloud Stack), dedicated servers, co-location, and managed services. PlusServer holds a BSI C5 attestation and explicitly

markets its offering to regulated industries in Germany and the broader EU, positioning data residency and operational transparency as core differentiators. It was selected for this study because of its explicit EU sovereignty positioning and its BSI C5 certification, which provides an independently verified compliance baseline that is directly relevant to the framework’s legal-control criteria.

### **T Cloud Public**

T Cloud Public is Deutsche Telekom’s purpose-built sovereign cloud platform, developed in parallel to the Open Telekom Cloud (OTC) as a direct strategic response to concerns about Huawei technology dependency [8]. Whereas OTC was co-developed with Huawei Technologies using Huawei FusionSphere as the core cloud operating system from 2016 onwards, T Cloud Public is designed from the outset as Deutsche Telekom’s sovereign-first response to the governance concerns surrounding OTC: it is operated exclusively by T-Systems EU engineers with Huawei prohibited from the production environment, is OpenStack-powered, and is approved for use by the German federal administration under the GovTech framework agreement for cloud and AI services [23]. Critically, both OTC and T Cloud Public continue to run on Huawei FusionSphere as their core cloud operating system — a dependency T-Systems acknowledges publicly — meaning T-Systems cannot independently maintain or update the foundational OS layer without Huawei’s continued software support. Both platforms operate from the same twin data centres in Biere and Magdeburg (Saxony-Anhalt, Germany), hold BSI C5 Type 2 attestation and ISO 27001 certification, and are managed exclusively by T-Systems EU-resident staff. T Cloud Public was included as a separate assessment target — alongside OTC — to capture Deutsche Telekom’s evolving sovereignty trajectory: OTC represents the legacy Huawei-dependent platform, while T Cloud Public represents the active sovereign remediation. This intra-provider comparison is itself a finding of the study and is discussed in Chapter 8.

## **4.2.2 France**

### **OVHcloud**

OVHcloud was founded in 1999 in Roubaix, France, by Octave Klaba, and completed an initial public offering on Euronext Paris in October 2021 [27]. It is Europe’s largest cloud provider measured by bare-metal server count and total data centre capacity, operating more than 40 data centres across Europe, North America, and Asia-Pacific, with the majority of its infrastructure anchored in France, Germany, and the United Kingdom. Its service catalogue spans bare-metal servers, public cloud IaaS, hosted private cloud (VMware), and a growing set of managed PaaS services. OVHcloud was selected as the pre-eminent European alternative to US hyperscalers: its scale, geographic reach, and investment in a broad managed-service portfolio make it the strongest single test case for whether a European provider can approach hyperscaler-level readiness.

### **Scaleway**

Scaleway was founded in 1999 as Online.net and rebranded under the Scaleway name in 2015 [29]. It is a wholly owned subsidiary of Iliad Group, the French telecommunications company known for its Free brand. Headquartered in Paris, Scaleway operates data centres in Paris, Amsterdam, and Warsaw, and powers its infrastructure with renewable energy. Its offering is developer-focused and spans bare-metal instances, managed Kubernetes, serverless functions, object storage, and a

growing portfolio of managed databases and networking services. Scaleway was selected as a French-origin IaaS provider with a maturing managed-service catalogue and a strong sustainability profile, representing the developer-community segment of the European cloud market.

### 4.2.3 Sweden

#### **Cleura**

Cleura was founded in 2009 under the name Citycloud and rebranded as Cleura in 2022 [5]. Headquartered in Gothenburg, Sweden, it is an OpenStack-native public cloud provider that emphasises data sovereignty, operational transparency, and operator accountability. Its infrastructure is located in Swedish data centres, and it offers compute, block and object storage, networking, and managed Kubernetes. Cleura distinguishes itself through its commitment to open-source infrastructure, customer data isolation, and clear contractual guarantees against third-party government access. It was selected as the Scandinavian OpenStack representative in this study and as a provider whose explicit data-sovereignty commitments can be directly mapped to the EC DG DIGIT framework criteria.

#### **Elastx**

Elastx was founded in 2013 and is headquartered in Gothenburg, Sweden [9]. It is an independent Swedish IaaS provider whose platform is built on OpenStack and powered entirely by renewable energy sourced from Nordic hydropower and wind. Its data centres are located in Sweden, and its service catalogue covers compute instances, block storage, object storage, and managed container services. As a smaller, bootstrapped provider, Elastx prioritises environmental sustainability, technical transparency, and a customer-first operational model. It was selected to represent the smaller-scale, sustainability-oriented segment of EU-native cloud providers and to test whether operational scale is a binding constraint on readiness and sovereignty scoring.

### 4.2.4 Switzerland

#### **Exoscale**

Exoscale was founded in 2012 in Lausanne, Switzerland, and is currently operated by A1 Digital International, a subsidiary of the A1 Telekom Austria Group [1]. Its data centres are located in Switzerland, Germany, Austria, and Bulgaria, giving it one of the broader multi-country footprints among mid-size European providers while maintaining governance under Swiss law. Its service catalogue focuses on IaaS compute, block and object storage, DNS, managed databases, and managed Kubernetes. Switzerland’s bilateral agreements with the EU and its robust Federal Act on Data Protection provide a distinct but closely aligned legal environment. Exoscale was selected for its multi-country EU footprint, its telecom-group ownership model, and the opportunity to examine how Swiss legal jurisdiction interacts with the EU sovereignty criteria.

#### **Infomaniak**

Founded in 1994 in Geneva, Switzerland, Infomaniak is an independent, employee-owned company with a long-standing privacy-first positioning [20]. It operates carbon-neutral data centres in Switzerland and complies with the Swiss Federal Act on Data Protection (nFADP), which shares

substantive principles with the GDPR. Its service portfolio spans web hosting, cloud servers, managed email and collaboration tools, and streaming infrastructure. Infomaniak has publicly committed to never transferring customer data to third countries without explicit consent and to resisting any extraterritorial data-access demands. It was selected as a long-standing, ethically oriented European provider that demonstrates how a small independent company can maintain a strong sovereignty posture through governance commitments rather than sheer infrastructure scale.

## **Nine**

Nine was founded in 2006 in Zurich, Switzerland, and operates as an independent Swiss hosting and cloud provider offering high-performance infrastructure and managed services [25]. Its data centres are located in Switzerland, and its service offering covers managed virtual machines, managed Kubernetes, databases, and professional-services engagements. Nine targets technology companies and digital agencies that require reliable, low-latency Swiss infrastructure with hands-on support. It was selected as a representative of the Swiss SME cloud segment, illustrating how specialised providers with a focused service catalogue position themselves within the broader European sovereign cloud ecosystem.

### **4.2.5 The Netherlands**

#### **Fuga Cloud (Cyso Group)**

Fuga Cloud is an OpenStack-based public cloud platform operated by Cyso Group, an Amsterdam-based Dutch IT services company [7]. Its infrastructure is located in Amsterdam, and its service catalogue covers compute, block and object storage, networking, and managed Kubernetes, with a community-oriented developer experience and transparent, usage-based pricing. Fuga Cloud maintains all data within the Netherlands and does not transfer data to non-EU jurisdictions. It was selected as the Dutch representative in this study and as an example of a community-oriented OpenStack provider whose governance model and geographic focus align closely with the framework’s operational-sovereignty criteria.

### **4.2.6 Finland**

#### **UpCloud**

UpCloud was founded in 2011 in Helsinki, Finland, and operates as a privately held Finnish cloud provider [31]. It is known for its MaxIOPS block storage architecture, which delivers high and consistent I/O performance, and for its low-latency European infrastructure. Its data centres span Helsinki, Frankfurt, Amsterdam, London, and Singapore, with the European nodes forming the primary customer-facing infrastructure. UpCloud offers IaaS compute, block and object storage, managed databases, managed Kubernetes, and networking services. It was selected for its performance-first positioning, its Nordic sovereignty context, and its status as a privately held Finnish company entirely independent of non-European ownership or investment.

### 4.2.7 On-Premises Reference

#### Locally Hosted Infrastructure

The locally hosted infrastructure profile represents a self-operated on-premises or private data centre deployment — an organisation’s own servers, networking, and storage managed by its own staff in facilities it physically controls. While not a commercial cloud provider, it is included in the study as a sovereignty reference point. It defines the theoretical ceiling for operational sovereignty: the organisation retains full physical control of hardware, full control of the software stack, and is exposed to no third-party terms of service, data-access agreements, or extraterritorial legal instruments that could compromise data confidentiality or availability. All commercial providers are necessarily compared against this ceiling, and the gap between a provider’s sovereignty score and the locally hosted baseline quantifies the sovereignty trade-off that accompanies the adoption of any third-party cloud service.

## 4.3 Intra-Provider Comparison: OTC and T Cloud Public

Two of the eighteen assessment targets — T-Systems Open Telekom Cloud (OTC) and T Cloud Public — share the same corporate parent (Deutsche Telekom AG / T-Systems International GmbH), the same physical data centres, and the same underlying cloud software stack. Their joint inclusion in this study is deliberate: they constitute a controlled intra-provider pair that isolates the effect of governance posture from technical capability, which is one of the central analytical claims of this thesis. Table 4.1 summarises their structural relationship.

Table 4.1: Structural comparison of T-Systems OTC and T Cloud Public

| Dimension               | T-Systems OTC                                 | T Cloud Public                                |
|-------------------------|-----------------------------------------------|-----------------------------------------------|
| Physical infrastructure | Biere and Magdeburg data centres (Germany)    | Same Biere and Magdeburg data centres         |
| Cloud engine            | Huawei FusionSphere (Open-Stack distribution) | Same Huawei FusionSphere                      |
| Service catalogue       | FunctionGraph, CCE, ECS, OBS, EVS, DCS...     | Same service names and catalogue              |
| Portal / URL            | open-telekom-cloud.com                        | public.t-cloud.com                            |
| Account and contract    | Separate commercial product                   | Separate commercial product                   |
| Launch and positioning  | 2016 — general-purpose public cloud           | 2023 — sovereign-positioned, GovTech-approved |
| Readiness (scalability) | 72%                                           | 74%                                           |
| Readiness (performance) | 87%                                           | 87%                                           |
| Sovereignty score       | 54.9%                                         | 65.9%                                         |

The two products are commercially distinct — a customer holding an OTC account does not automatically have access to T Cloud Public, and their contractual terms, support tiers, and pricing structures differ. Yet because they share the same Huawei FusionSphere cloud operating system, the same data centre facilities, and the same service catalogue (FunctionGraph for serverless, Cloud Container Engine for Kubernetes, Elastic Cloud Server for compute, Object Storage Service for blob

storage, and so on), their readiness scores are nearly identical. The marginal gap in scalability (72 vs 74) is attributable to T Cloud Public’s additional Netherlands capacity and its GovTech-approved infrastructure designation, not to any difference in deployed services.

Despite this technical parity, the sovereignty scores diverge by 11 percentage points (54.9% for OTC; 65.9% for T Cloud Public). The differences are not artefacts of inconsistent scoring; they reflect deliberate governance choices that T-Systems made when designing T Cloud Public as a sovereign-first product in 2023, distinct from the 2016 architecture decisions that shaped OTC. The principal divergences across the EC DG DIGIT framework objectives are as follows.

**SOV-1 (Strategic Sovereignty):** OTC scores 50% against T Cloud Public’s 83%. Three questions drive the gap. On decisive authority (Q1), OTC receives *Partial* because Huawei FusionSphere gives a Chinese-controlled entity technical authority over OTC’s platform OS, whereas T Cloud Public receives *Yes* because T-Systems operates the platform entirely independently and Huawei has zero production system access. On EU financing (Q3), T Cloud Public receives *Yes* owing to IPCEI-CIS public co-financing and KfW as the primary shareholder; OTC receives *Partial* because its financing is not tied to EU sovereign investment instruments. On EU sovereignty-initiative engagement (Q5), T Cloud Public receives *Yes* by virtue of its May 2026 GovTech Deutschland procurement approval for German federal administration; OTC receives *Partial* because, despite Deutsche Telekom’s GAIA-X founding membership, OTC was co-developed with Huawei and was never designed to meet EU sovereign cloud criteria.

**SOV-2 (Legal and Jurisdictional):** OTC scores 40% against T Cloud Public’s 50%. The gap is attributable to Q3 (non-EU data-access channels), where OTC receives *No*: Huawei engineers participate in OTC’s third-level support escalations and the Huawei FusionSphere layer cannot be verified as free of capabilities invocable under China’s National Intelligence Law. T Cloud Public receives *Partial* on the same question because, while the same theoretical Huawei supply-chain risk applies, T-Systems enforces a zero-access policy for Huawei personnel in production and documents this as a formal operational control.

**SOV-4 (Operational Sovereignty):** OTC scores 50% against T Cloud Public’s 67%. On full EU-operator management (Q2), OTC receives *No*: Huawei engineers participate in OTC third-level support via video conference with no formally documented zero-access constraint, and OTC has no alternative OS strategy if Huawei withdraws software support. T Cloud Public receives *Partial*: the same FusionSphere OS dependency exists, but Huawei consultation is explicitly zero-access and video-only, constituting a materially stronger procedural control. On EU-exclusive support delivery (Q4), OTC receives *Partial* and T Cloud Public receives *Yes*, reflecting the same distinction: T Cloud Public’s operational model formally bars any non-EU support routing at all levels.

**SOV-5 (Supply Chain):** OTC scores 20% against T Cloud Public’s 30%. The sole difference is hardware manufacturing (Q1): T Cloud Public receives *Partial* because its dual-vendor hardware strategy has qualified western server chassis (Intel Xeon 6900 CPUs in non-Huawei enclosures for c9 instances, NVIDIA DGX for the Industrial AI Cloud), representing a concrete trajectory toward trusted-jurisdiction hardware. OTC receives *No* because no equivalent diversification programme exists and Deutsche Telekom’s stated strategy is to migrate OTC customers to T Cloud Public rather than remediate OTC’s supply chain.

**SOV-8 (Environmental):** OTC scores 75% against T Cloud Public’s 88%. Two questions drive the gap. On Q1 (energy efficiency and PUE), both providers receive *Partial*: the Biere and Magdeburg data centres publish quarterly PUE figures of approximately 1.28–1.3, which exceed the  $\leq 1.2$  threshold required for a *Yes* verdict. On Q3 (carbon and water disclosures), T Cloud Public receives *Yes* because Deutsche Telekom’s 2024 Corporate Responsibility Report (externally assured,

CSRD-compliant) includes per-data-centre PUE figures and full Scope 1–3 GHG disclosures; OTC receives *Partial* because OTC-specific Water Usage Effectiveness metrics are not published at the product level.

The pattern across these divergences confirms that T Cloud Public’s sovereignty advantage over OTC is entirely governance-procedural. No new hardware was installed, no new software was deployed, and no new data centres were built to achieve the 11-point score improvement. The gain came exclusively from formalised access controls, documented review processes, EU-exclusive support policies, EU sovereign investment participation, and government procurement approval — decisions made at the product governance layer. This finding is developed further in the discussion in Chapter 8, where it is used to substantiate the thesis argument that *cloud sovereignty is an outcome of governance posture, not infrastructure capability*.

## 4.4 AWS as Reference Baseline

Amazon Web Services (AWS) was included in this study not as a selected European provider but as the universal readiness baseline against which all other providers are evaluated. Launched in 2006 and consistently ranked as the global cloud market leader [17], AWS offers the most comprehensive commercially available cloud service catalogue, spanning compute, storage, networking, security, managed databases, AI and machine learning, developer tooling, analytics, and IoT services. This breadth makes it the natural reference for the Cloud Readiness Assessment described in Chapter 5. In that assessment, AWS is assigned a readiness score of 100%, representing full coverage of all 60 evaluated service categories. Each European provider is then scored relative to this benchmark, and the resulting percentage reflects how completely it could substitute for AWS in a typical enterprise workload. The AWS baseline does not imply that AWS is a recommended or sovereignty-compatible choice for EU organisations; on the contrary, its jurisdictional exposure under the CLOUD Act and FISA Section 702 is precisely the driver of the sovereignty deficit that this thesis measures.

## 4.5 Summary

Table 4.2 provides a consolidated overview of all 18 providers, including their country of registration, founding year, ownership structure, and primary cloud service tier.

The selection spans six countries — Germany, France, Sweden, Switzerland, the Netherlands, and Finland — and encompasses a diverse range of ownership models, including family-owned independents, publicly listed technology companies, telecom-group subsidiaries, and conglomerate-backed new entrants. Notably, two of the 18 entries — T-Systems OTC and T Cloud Public — represent different platforms from the same corporate parent (Deutsche Telekom / T-Systems), included separately to capture the provider’s active sovereignty transition: OTC is the legacy Huawei-dependent platform, while T Cloud Public is its purpose-built sovereign successor. This intra-provider pair is itself an analytical finding, demonstrating that a single organisation can occupy meaningfully different positions on the sovereignty–readiness frontier depending on which platform generation is assessed. The inclusion of locally hosted infrastructure as an eighteenth data point anchors both the sovereignty ceiling and the readiness floor, providing a complete reference span against which all commercial providers can be positioned.

Table 4.2: Summary of selected EU cloud providers

| Provider                      | Country     | Founded | Parent / Ownership                         | Cloud Tier |
|-------------------------------|-------------|---------|--------------------------------------------|------------|
| Hetzner                       | Germany     | 1997    | Independent (family-owned)                 | IaaS       |
| IONOS                         | Germany     | 1988    | United Internet AG                         | IaaS/PaaS  |
| STACKIT                       | Germany     | 2021    | Schwarz Group                              | IaaS/PaaS  |
| T-Systems Open Telekom Cloud  | Germany     | 2016    | Deutsche Telekom AG / T-Systems            | IaaS/PaaS  |
| T Cloud Public                | Germany     | 2023    | Deutsche Telekom AG / T-Systems            | IaaS/PaaS  |
| Arvato Systems                | Germany     | 1961    | Bertelsmann SE & Co. KGaA                  | Managed    |
| Noris Network                 | Germany     | 1993    | Independent                                | Managed    |
| PlusServer                    | Germany     | 1999    | Ioniq Group                                | IaaS/PaaS  |
| OVHcloud                      | France      | 1999    | Independent (publicly listed, Kaba family) | IaaS/PaaS  |
| Scaleway                      | France      | 1999    | Iliad Group                                | IaaS/PaaS  |
| Cleura                        | Sweden      | 2009    | Independent                                | IaaS       |
| Elastx                        | Sweden      | 2013    | Independent                                | IaaS       |
| Exoscale                      | Switzerland | 2012    | A1 Telekom Austria Group                   | IaaS/PaaS  |
| Infomaniak                    | Switzerland | 1994    | Independent (employee-owned)               | IaaS/PaaS  |
| Nine                          | Switzerland | 2006    | Independent                                | Managed    |
| Fuga Cloud (Cyso Group)       | Netherlands | 2013    | Cyso Group (independent)                   | IaaS       |
| UpCloud                       | Finland     | 2011    | Independent                                | IaaS/PaaS  |
| Locally Hosted Infrastructure | N/A         | N/A     | Organisation-owned                         | On-Prem    |

# Chapter 5

## Methodology

### 5.1 Research Design and Overall Approach

This thesis adopts a constructive research design, combining framework-driven assessment with empirical tool development. Rather than observing cloud sovereignty as a static phenomenon, the research operationalises it through two complementary instruments that can be applied, reproduced, and extended by practitioners. The first instrument — the Sovereignty Scoring Framework (SovScore) — measures the degree of digital sovereignty a cloud provider offers to EU customers. The second — the Cloud Readiness Assessment — measures how prepared each provider is to substitute AWS-equivalent managed services for a European enterprise workload. Both instruments are grounded in publicly available regulatory documents and evaluated against a curated set of 18 European cloud providers.

The two experiments are described in detail in Sections 5.3 and 5.4 respectively. Both were conducted using an open-source web tool developed as part of this research, described in the following section.

### 5.2 CloudSov Tool: Implementation and Usage

#### 5.2.1 Overview and Availability

To ensure reproducibility and accessibility, both experiments were operationalised within a dedicated open-source web application named **CloudSov**. The tool is publicly available on GitHub and can be run locally without a cloud account or paid subscription. It provides a graphical interface for conducting the SovScore assessment and the Cloud Readiness Assessment in a structured, comparable manner across all 18 EU providers.

*Repository:* <https://github.com/aaeu-ihono/cloudsov-tool.git> *Licence:* MIT

The tool is built with a Python/FastAPI backend and a React 19 (Vite) frontend, communicating over a local REST API. Provider responses are stored as structured JSON files that can be audited independently of the interface.

## 5.2.2 Getting Started

Prerequisites: Python 3.10+, Node.js 18+, and npm. To launch the tool:

1. Clone the repository: `git clone https://github.com/aaeu-ihono/cloudsov-tool.git`
2. Start the backend: `cd backend && pip install -r requirements.txt && uvicorn main:app --reload`
3. Start the frontend: `cd frontend && npm install && npm run dev`
4. Open a browser at `http://localhost:5173`

No environment variables or external API keys are required. All provider data is bundled with the repository under the `survey/` and `readiness/` directories.

## 5.2.3 Interface Walkthrough

The tool interface is divided into two primary modules, accessible from the top navigation bar: **SovScore** and **Readiness Assessment**. Figures 5.1–5.3 illustrate each section. The numbered annotations in the figures correspond to the legend in Table 5.1.

Table 5.1: Interface annotation legend — numbered regions shown in Figures 5.1–5.3

| No. | Description                                                                                                          |
|-----|----------------------------------------------------------------------------------------------------------------------|
| ①   | Provider selector and add/remove controls — manage which providers appear in the assessment grid                     |
| ②   | SOV objective column headers — each column represents one of the eight sovereignty objectives (SOV-1 to SOV-8)       |
| ③   | Verdict input cells — each cell accepts Y (Yes), P (Partial), or N (No) for the corresponding provider-question pair |
| ④   | Computed SovScore column — displays the weighted aggregate score (0–100) updated in real time                        |
| ⑤   | SEAL badge — colour-coded maturity level (SEAL 0 to SEAL 4) derived from the score                                   |
| ⑥   | Dimension breakdown radar/bar chart — shows per-dimension readiness scores for a selected provider                   |
| ⑦   | Provider toggle panel — enables or disables individual providers on the readiness line chart                         |
| ⑧   | Gap table — lists all 60 services ranked by average gap from AWS baseline; expandable rows show per-provider status  |

To use the **SovScore** module (Figure 5.1): select the providers to compare using the panel at the top ①, then work through each column ②, entering verdicts in the cells ③. The score ④ and SEAL badge ⑤ update automatically. Hovering over a cell reveals the full question text and an evidence note field where source URLs can be recorded.

To use the **Readiness Assessment** module (Figure 5.2): toggle providers on or off using the pill buttons ⑦. The line chart displays dimension scores as a time-series of 13 capability dimensions. Hovering over any data point reveals a tooltip listing which of the 60 services are available (Y),

partial (P), or missing (N) for that provider in that dimension. The gap table ⑧ (Figure 5.3) ranks services by average deficit and allows drill-down per provider.

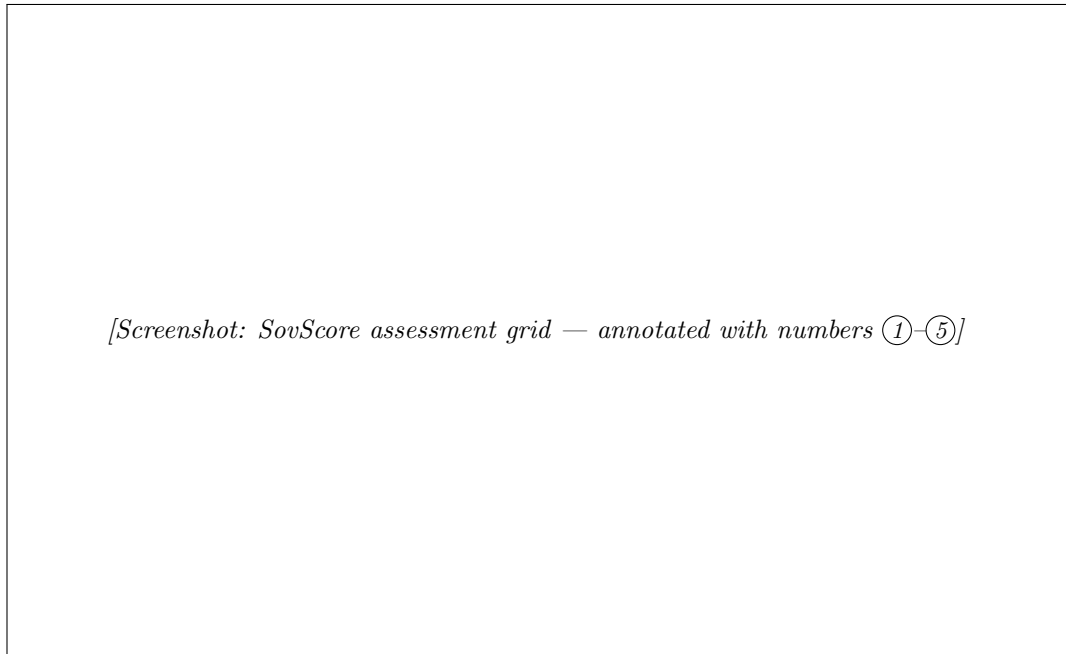


Figure 5.1: CloudSov SovScore module: provider grid with verdict inputs and computed scores.

#### 5.2.4 Advantages, Limitations, and Controversies

**Advantages.** CloudSov addresses the absence of a unified, comparable, open evaluation framework for EU cloud sovereignty. By combining two complementary instruments in a single interface, it enables side-by-side comparison across providers that would otherwise require bespoke spreadsheets and manual aggregation. The tool is technology-neutral, provider-agnostic, and entirely reproducible: any researcher with access to the repository can re-run or extend the assessment.

**Limitations.** The assessments rely on publicly available documentation and self-reported provider claims. No independent technical audit of the verdicts was conducted. Providers with sparse public documentation may receive lower scores not because they fail to meet the criteria, but because sufficient evidence could not be established — a documentation gap rather than a capability gap. Furthermore, the SovScore weights are prescribed by the EC DG DIGIT framework and were not derived empirically; alternative weighting schemes would yield different rankings.

**Controversies.** The use of AWS as a readiness baseline has inherent tensions: it implicitly treats the US hyperscaler’s service model as the normative standard against which EU alternatives are measured, which may undervalue architectural choices made by EU providers (e.g., OpenStack-based deployments that prioritise interoperability over proprietary services). Additionally, the

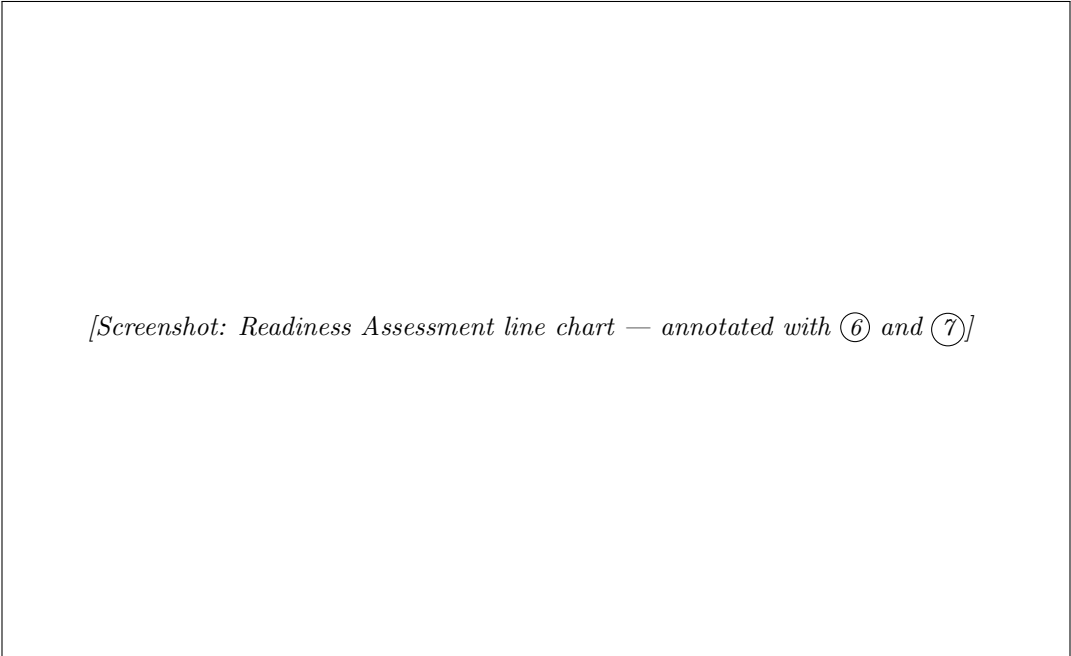


Figure 5.2: CloudSov Readiness Assessment module: dimension scores across providers.

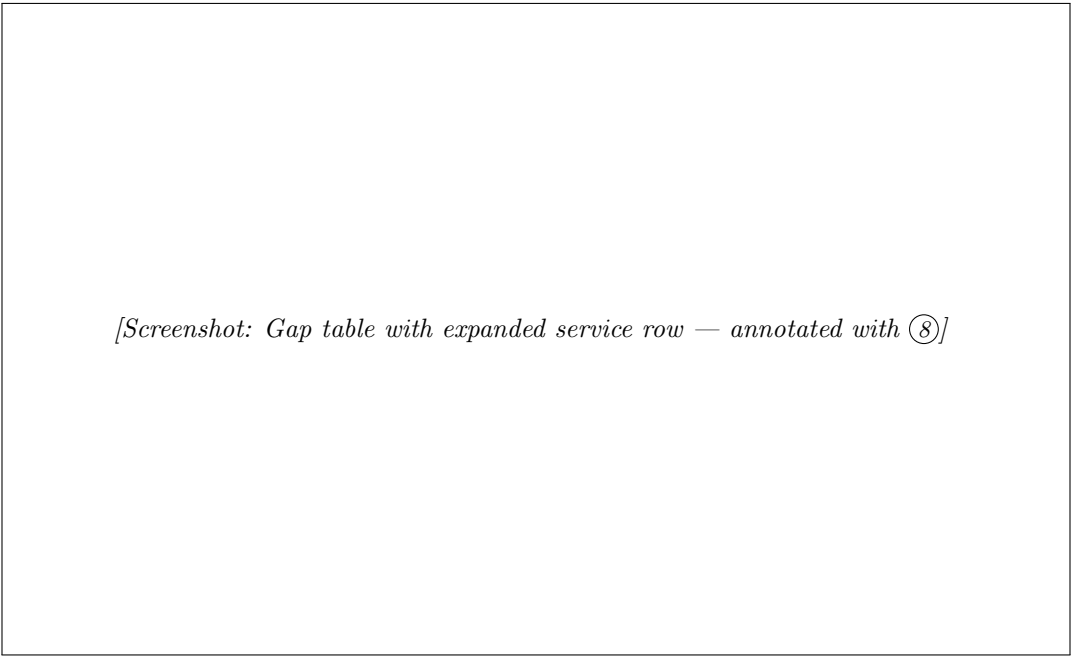


Figure 5.3: CloudSov gap table: services ranked by average deficit from the AWS baseline.

binary gap model (available / partial / missing) cannot capture quality, performance, or SLA differences between an EU provider’s service and its AWS equivalent. These limitations are acknowledged in the discussion of findings in Chapter 8.

## 5.3 Sovereignty Scoring Framework (SovScore)

### 5.3.1 Theoretical Basis and Framework Selection

Cloud computing has historically been evaluated along performance, reliability, and cost dimensions [3, 24]. The policy dimension — concerning legal jurisdiction, data control, and strategic independence from non-EU actors — has remained largely qualitative and inconsistently defined across the literature. This experiment addresses that gap by grounding the sovereignty assessment in the most authoritative publicly available specification for EU cloud governance: the European Commission Directorate-General for Digital Services *Cloud Sovereignty Framework v1.2.1*, published in October 2025 [13].

The decision to adopt the EC DG DIGIT framework over alternative sovereign cloud criteria<sup>1</sup> was motivated by three factors. First, it is produced by the Commission body responsible for EU digital policy, giving it direct regulatory standing. Second, it decomposes sovereignty into eight thematic objectives with explicit sub-criteria and a five-level maturity scale. Third, it is technology-neutral and provider-agnostic, making it applicable across the heterogeneous landscape of European cloud providers without presupposing any particular architectural pattern.

### 5.3.2 SOV Objectives and Weighting Scheme

The framework defines eight *SOV objectives*, each representing a distinct dimension of digital sovereignty. Each objective is assigned a percentage weight reflecting its relative importance to overall sovereignty, as specified in the EC DG DIGIT framework. These weights sum to 100% and are held constant across all provider assessments to ensure comparability.

Table 5.2 summarises the eight objectives with their weights and minimum SEAL thresholds. Tables 5.3 and 5.4 list the full evaluation questions for each objective.

The weights reflect the prioritisation established by the EC DG DIGIT framework. Supply Chain (SOV-5, 20%) carries the highest weight, recognising that hardware and software dependencies represent the most structurally embedded sovereignty risk. Strategic Sovereignty (SOV-1) and Operational Sovereignty (SOV-4) jointly account for 30%, reflecting the importance of EU-controlled governance and operational independence. Environmental performance (SOV-8, 5%) carries the lowest weight, as it is not a sovereignty disqualifier under current EU regulation, though it is tracked for alignment with the EU Green Deal and the Corporate Sustainability Reporting Directive (CSRD).

---

<sup>1</sup>Alternative frameworks considered include the ENISA EUCS certification scheme [16] and the GAIA-X Trust Framework [18]. The EC DG DIGIT document was preferred because it provides explicit, question-level evaluation criteria that map directly to measurable provider attributes, rather than certification labels that presuppose a third-party audit process.

Table 5.2: SOV objectives, weights, question counts, and minimum SEAL thresholds

| ID           | Objective               | Weight (%) | Questions | Min. SEAL |
|--------------|-------------------------|------------|-----------|-----------|
| SOV-1        | Strategic Sovereignty   | 15         | 6         | 3         |
| SOV-2        | Legal & Jurisdictional  | 10         | 5         | 4         |
| SOV-3        | Data & AI Sovereignty   | 10         | 5         | 3         |
| SOV-4        | Operational Sovereignty | 15         | 6         | 2         |
| SOV-5        | Supply Chain            | 20         | 5         | 2         |
| SOV-6        | Technology              | 15         | 4         | 2         |
| SOV-7        | Security & Compliance   | 10         | 6         | 3         |
| SOV-8        | Environmental           | 5          | 4         | 1         |
| <b>Total</b> |                         | <b>100</b> | <b>41</b> |           |

Table 5.3: SOV objectives SOV-1 to SOV-4: weights, min. SEAL thresholds, and evaluation questions

|    | <b>SOV-1</b><br>Strategic Sovereignty<br><i>15% · min SEAL-3</i> | <b>SOV-2</b><br>Legal & Jurisdictional<br><i>10% · min SEAL-4</i> | <b>SOV-3</b><br>Data & AI Sovereignty<br><i>10% · min SEAL-3</i> | <b>SOV-4</b><br>Operational Sovereignty<br><i>15% · min SEAL-2</i> |
|----|------------------------------------------------------------------|-------------------------------------------------------------------|------------------------------------------------------------------|--------------------------------------------------------------------|
| Q1 | Decisive authority bodies within EU?                             | Governed exclusively by EU member law?                            | Customer-managed encryption keys (CMEK)?                         | Workload portability without vendor lock-in?                       |
| Q2 | Structural protection against non-EU takeover?                   | Free from CLOUD Act and extraterritorial laws?                    | Full auditability of data and AI access?                         | EU-managed operations, no non-EU vendor dependency?                |
| Q3 | Primarily EU-sourced financing?                                  | No channel for non-EU compelled data access?                      | Verifiable irreversible data deletion?                           | EU-based talent pool for operations?                               |
| Q4 | Significant EU investment and job creation?                      | Free from non-EU export control regimes?                          | Data storage and processing EU-only?                             | All operational support from within EU?                            |
| Q5 | Active in GAIA-X and EU sovereignty initiatives?                 | IP registered and developed within EU?                            | AI models trained under EU control?                              | Technical documentation and source code available?                 |
| Q6 | Operational continuity if non-EU support withdrawn?              | —                                                                 | —                                                                | All critical suppliers under EU jurisdiction?                      |

Table 5.4: SOV objectives SOV-5 to SOV-8: weights, min. SEAL thresholds, and evaluation questions

|    | <b>SOV-5</b><br>Supply Chain<br><i>20% · min SEAL-2</i> | <b>SOV-6</b><br>Technology<br><i>15% · min SEAL-2</i> | <b>SOV-7</b><br>Security & Compliance<br><i>10% · min SEAL-3</i> | <b>SOV-8</b><br>Environmental<br><i>5% · min SEAL-1</i> |
|----|---------------------------------------------------------|-------------------------------------------------------|------------------------------------------------------------------|---------------------------------------------------------|
| Q1 | Hardware manufactured in EU or trusted jurisdictions?   | Non-proprietary APIs and open standards?              | Recognised EU certifications (ISO 27001, BSI C5)?                | Energy-efficient infrastructure with low PUE?           |
| Q2 | Firmware of EU origin?                                  | Open-source licensed software stack?                  | Compliant with GDPR, NIS2, and DORA?                             | Circular economy practices for hardware?                |
| Q3 | Software stack architected and updated within EU?       | Full architectural transparency?                      | All SOCs operating under EU jurisdiction?                        | Transparent carbon and water disclosure?                |
| Q4 | No significant reliance on non-EU vendors?              | Free from non-EU HPC or processor dependency?         | Customer oversight of security monitoring?                       | Renewable or low-carbon energy sourcing?                |
| Q5 | Full supply chain transparency and audit rights?        | —                                                     | EU-compliant breach reporting and patch management?              | —                                                       |
| Q6 | —                                                       | —                                                     | Independent EU security audit rights?                            | —                                                       |

### 5.3.3 Evidence Collection: The Y/P/N Verdict Protocol

For each of the 41 questions across the eight SOV objectives, a verdict is assigned per provider based on publicly available evidence. Three verdict values are used:

- **Y (Yes)** — the provider fully and verifiably meets the criterion.
- **P (Partial)** — the provider partially meets the criterion, with documented limitations or caveats.
- **N (No)** — the provider does not meet the criterion, or no credible evidence of compliance was found.

Evidence sources include official provider documentation, annual reports, third-party certification records, press releases, and publicly disclosed regulatory filings. Each verdict is accompanied by a source URL and an explanatory note stored in a structured JSON survey file per provider. The evidence collection process was conducted systematically across all 18 providers using the same set of 41 questions, ensuring comparability.

Where a criterion cannot be assessed from public information alone, the conservative verdict N is applied, consistent with the principle that sovereignty claims must be affirmatively demonstrable rather than assumed by default.<sup>2</sup>

<sup>2</sup>This is particularly relevant for SOV-5 (Supply Chain), where hardware firmware provenance is rarely disclosed publicly. Intel Management Engine and AMD PSP firmware, present in virtually all commercial x86 servers, are of non-EU origin, creating a structural ceiling that applies uniformly across all providers assessed.

### 5.3.4 SEAL Level Computation

Each verdict is converted to a numeric score (Y = 2, P = 1, N = 0). The raw score for a provider on a given SOV objective is the sum of its verdict scores across that objective’s questions. This is mapped to a *Sovereignty Assurance Level* (SEAL) on a five-point scale (0–4) using the thresholds defined in Table 5.5.

Table 5.5: SEAL level thresholds and descriptions

| SEAL | Name                       | Description                                                                           |
|------|----------------------------|---------------------------------------------------------------------------------------|
| 0    | No Sovereignty             | Service under exclusive control of non-EU third parties.                              |
| 1    | Jurisdictional Sovereignty | EU law formally applies with limited practical enforceability.                        |
| 2    | Data Sovereignty           | EU law applicable and enforceable; material non-EU dependencies remain.               |
| 3    | Digital Resilience         | EU law enforceable; EU actors exercise meaningful but not full influence.             |
| 4    | Full Digital Sovereignty   | Technology and operations under complete EU control; no critical non-EU dependencies. |

The mapping from raw score to SEAL level uses fixed percentage thresholds applied to the maximum achievable score for each objective:

$$SEAL_i = \begin{cases} 0 & \text{if } p_i \leq 0.20 \\ 1 & \text{if } 0.20 < p_i \leq 0.40 \\ 2 & \text{if } 0.40 < p_i \leq 0.60 \\ 3 & \text{if } 0.60 < p_i \leq 0.80 \\ 4 & \text{if } p_i > 0.80 \end{cases} \quad \text{where } p_i = \frac{\text{score}_i}{\text{max\_score}_i} \quad (5.1)$$

The overall *sovereignty score* for a provider is then computed as a weighted sum of normalised SEAL levels across all eight objectives:

$$S_{\text{sov}} = \sum_{i=1}^8 \frac{SEAL_i}{4} \times w_i \times 100 \quad (\%) \quad (5.2)$$

where  $w_i$  is the weight of objective  $i$  (as listed in Table 5.2) and dividing by 4 normalises each SEAL to the  $[0, 1]$  interval. The resulting score  $S_{\text{sov}}$  lies in the range  $[0, 100]\%$ , where 100% corresponds to a hypothetical provider achieving SEAL-4 on all eight objectives.

### 5.3.5 Minimum SEAL Thresholds and Pass/Fail Determination

A sovereignty score alone does not determine whether a provider is fit for use in a regulated enterprise context. Each SOV objective carries a minimum acceptable SEAL level (see Table 5.2), below which the provider is considered non-compliant on that objective regardless of its scores on other objectives. A provider *passes* the assessment only if it meets or exceeds the minimum SEAL threshold on every objective simultaneously.

The minimum thresholds were set based on the regulatory environment applicable to enterprise cloud procurement in the EU. For instance, SOV-2 (Legal & Jurisdictional) requires a minimum of SEAL-4, reflecting the binary nature of legal jurisdiction: a single residual channel through which a non-EU authority could compel data disclosure — such as the US CLOUD Act<sup>3</sup> or the Chinese Cybersecurity Law — disqualifies a provider from processing personal data under GDPR [14]. SOV-5 (Supply Chain) is set at SEAL-2, acknowledging the structural impossibility of full hardware sovereignty given the global x86 supply chain.

### 5.3.6 Provider Coverage and Tool Implementation

The assessment was applied to 18 European cloud providers selected on the basis of EU headquarters, commercial availability, and relevance to enterprise workloads. For each provider, a structured JSON survey file was compiled containing the 41 verdicts with associated source URLs and explanatory notes.

The scoring framework was implemented as an interactive web application (SovScore) built with React 19 and FastAPI, allowing real-time exploration of results, adjustment of minimum SEAL thresholds, and comparison across providers. The application computes  $S_{\text{sov}}$  from equation (5.2) in the browser on every state change, with all verdict data served statically from the JSON survey files.

*[SCREENSHOT REQUIRED: SovScore web interface showing the full provider  $\times$  question grid, SEAL chips per SOV objective, and the Score column with sort order applied. Capture with all 18 providers visible and at least one provider highlighted in red (failing).]*

Figure 5.4: SovScore web interface: provider assessment grid with SEAL levels and overall sovereignty scores.

*[SCREENSHOT REQUIRED: Hover tooltip on a SOV objective header (e.g. SOV-2 Legal & Jurisdictional), showing the ⓘ popover with the min-SEAL justification text.]*

Figure 5.5: Min-SEAL justification tooltip on the SovScore grid header.

<sup>3</sup>The Clarifying Lawful Overseas Use of Data (CLOUD) Act (2018) allows US law enforcement to compel US-headquartered companies to produce data stored outside the United States, irrespective of the storage location. Its incompatibility with GDPR was confirmed by the CJEU in *Schrems II* [6].

### 5.3.7 Preliminary SovScore Findings

The SovScore assessment surfaces a striking pattern across the 18 providers: the relationship between certification depth and sovereignty score is not linear. Providers holding the most extensive EU compliance portfolios — notably T-Systems (T Cloud Public) with BSI C5 Type 2, ISO 27001, SOC 1/2/3, TISAX, and GDPR Art. 28 DPAs — score near-perfectly on Security & Compliance (SOV-7) and Environmental (SOV-8) dimensions. Yet the same provider records the study’s most complex sovereign weakness: a persistent partial score on Legal & Jurisdictional (SOV-2) and a failed verdict on Supply Chain (SOV-5) due to the Huawei FusionSphere dependency inherited from the original Open Telekom Cloud architecture. China’s National Intelligence Law (2017) theoretically obligates Chinese companies to cooperate with state intelligence services, creating a residual extraterritorial risk that even the most rigorous German certification cannot fully eliminate.

A second observation is the universal failure of all EU providers on the Technology dimension (SOV-6) question regarding EU-origin high-performance computing hardware. No commercially available EU cloud provider deploys EU-manufactured server processors or GPU accelerators at scale; all depend on Intel or AMD x86 silicon (US origin) and NVIDIA accelerators (US origin). This is a structural constraint of the current semiconductor industry rather than a provider-specific deficiency, and it establishes a ceiling below which any EU cloud provider’s SOV-6 score must fall regardless of governance choices.

A full cross-provider analysis of SovScore results, including ranked scores, SEAL-level breakdowns per objective, and pass/fail determinations against the minimum thresholds, is presented in Chapter 8.

## 5.4 Cloud Readiness Assessment Design

### 5.4.1 Rationale and Baseline Selection

Sovereignty scoring measures whether a provider is legally and structurally safe to use under EU jurisdiction. It does not, however, measure whether a provider can actually replace the managed services that an enterprise already depends on. A cloud provider may achieve a high sovereignty score while lacking several services critical to a given workload, creating significant migration friction. The Cloud Readiness Assessment addresses this complementary question.

Amazon Web Services was selected as the 100% capability baseline for three reasons. First, AWS holds the largest share of the global enterprise cloud market and is the de facto incumbent for the majority of European enterprises currently operating in a hyperscaler environment [3]. Second, AWS offers the most comprehensive publicly documented managed service portfolio, providing a stable reference taxonomy. Third, the use case driving this research — evaluating migration feasibility for Alps Alpine, operating on AWS — makes AWS the natural “as-is” baseline from which a transition to a European provider would depart.

The assessment is not a judgment on AWS’s technical quality; rather, it uses AWS’s service portfolio as a reference map of enterprise cloud capability requirements against which European alternatives are measured.

### 5.4.2 Service Taxonomy: 11 Categories and 60 Services

Sixty AWS managed services were selected as the evaluation rubric, organised into 11 functional categories. The selection was guided by two criteria: (1) services identified as required in the Alps Alpine IT infrastructure review, and (2) services appearing consistently in enterprise cloud reference architectures across the industry, as evidenced by the AWS Well-Architected Framework [2] and confirmed by enterprise adoption data in the Flexera State of the Cloud Report [17]. Services meeting both criteria are classified as *required*; services meeting only the second are included as *industry-standard additions* and are annotated accordingly in the tool.

Table 5.6 lists all 60 services across the 13 assessment dimensions. Services marked <sup>†</sup> are industry-standard inclusions not specifically identified as required by Alps Alpine but included to ensure full coverage of enterprise cloud reference architectures [2, 17].

Table 5.6: All 60 AWS services assessed across 13 readiness dimensions (<sup>†</sup> = industry-standard; not specifically required by Alps Alpine)

| Scalability         | Performance             | Application (1) | Compute (6)                    | Storage (5)      | Database (7)             | Networking (7)              |
|---------------------|-------------------------|-----------------|--------------------------------|------------------|--------------------------|-----------------------------|
| <i>Pre-assessed</i> | <i>Pre-assessed</i>     | Amplify         | Lambda                         | S3               | RDS                      | App. Load Balancer          |
|                     |                         |                 | ECS                            | S3 Glacier       | DocumentDB               | Net. Load Balancer          |
|                     |                         |                 | EKS                            | EBS              | DynamoDB                 | VPC                         |
|                     |                         |                 | EC2                            | Backup           | Timestream               | Route 53                    |
|                     |                         |                 | Batch                          | EFS <sup>†</sup> | Aurora <sup>†</sup>      | API Gateway                 |
|                     |                         |                 | Elastic Beanstalk <sup>†</sup> |                  | ElastiCache <sup>†</sup> | CloudFront <sup>†</sup>     |
|                     |                         |                 |                                |                  | Redshift <sup>†</sup>    | Direct Connect <sup>†</sup> |
| AI/ML (6)           | Security (12)           | Monitoring (4)  | Dev/Deploy (5)                 | IoT (2)          | Messaging (5)            |                             |
| Rekognition         | IAM                     | CloudWatch      | CodeBuild                      | IoT Core         | SQS                      |                             |
| SageMaker           | Cognito                 | SNS             | CodeDeploy                     | IoT Device Mgmt. | EventBridge              |                             |
| Bedrock             | IAM Identity Center     | Systems Manager | CodePipeline                   |                  | Step Functions           |                             |
| Glue                | KMS                     | Cost Explorer   | ECR                            |                  | SES                      |                             |
| EMR                 | Secrets Manager         |                 | CloudFormation <sup>†</sup>    |                  | Kinesis <sup>†</sup>     |                             |
| Athena              | Certificate Manager     |                 |                                |                  |                          |                             |
|                     | GuardDuty               |                 |                                |                  |                          |                             |
|                     | Security Hub            |                 |                                |                  |                          |                             |
|                     | Config                  |                 |                                |                  |                          |                             |
|                     | Shield                  |                 |                                |                  |                          |                             |
|                     | WAF                     |                 |                                |                  |                          |                             |
|                     | CloudTrail <sup>†</sup> |                 |                                |                  |                          |                             |

### 5.4.3 Coverage Scoring Protocol

For each of the 60 services, every EU provider under assessment receives a *coverage verdict* indicating whether a functionally equivalent managed service exists in their portfolio:

- **Y (Full)** — a managed equivalent service exists with comparable functionality and general availability status.
- **P (Partial)** — an equivalent exists but is in beta, feature-limited, or requires a workaround that increases operational overhead.
- **N (None)** — no managed equivalent is available; self-hosting on IaaS would be required, introducing additional operational burden.

This three-level coverage scheme was deliberately aligned with the Y/P/N verdict protocol used in the SovScore experiment, allowing a consistent scoring scale across both instruments. Where a partial equivalent exists, the EU service name or approach is recorded alongside the verdict to provide actionable migration guidance.

#### 5.4.4 Dimension Score Computation

Coverage verdicts are converted to points ( $Y = 2$ ,  $P = 1$ ,  $N = 0$ ). The score for a provider on a given feature dimension  $d$  is:

$$R_d = \frac{\sum_{s \in d} \text{cov}(s)}{2 \times |d|} \times 100 \quad (\%) \quad (5.3)$$

where  $|d|$  is the number of services in dimension  $d$  and  $\text{cov}(s) \in \{0, 1, 2\}$  is the coverage point value for service  $s$ . A score of 100% indicates full managed-service parity with AWS on that dimension; 0% indicates no managed equivalents exist and all services would require self-hosting.

Scalability and Performance are assessed separately as pre-scored dimensions (0–100) derived from published regional availability, auto-scaling capability documentation, and official SLA commitments, rather than from service-level verdicts.

The average readiness score across all 13 dimensions is used to rank providers and compute the *average gap to AWS* as:

$$\text{Gap} = \frac{1}{13} \sum_{d=1}^{13} (100 - R_d) \quad (\text{percentage points}) \quad (5.4)$$

A smaller gap indicates higher overall feature parity with AWS and therefore lower expected migration friction.

#### 5.4.5 Tool Implementation and Visualisation

The Cloud Readiness Assessment was implemented as the second module of the CloudSov web application, co-located with the SovScore tool. The frontend presents three complementary views of the assessment data:

1. A **radar chart** showing each provider’s coverage profile across all 13 dimensions simultaneously, with the AWS reference polygon as the outer boundary.
2. An **area chart** showing dimension-by-dimension coverage with per-provider lines, enabling identification of specific capability gaps. Hovering over a data point reveals the coverage verdict for each service in that dimension.
3. A **ranked gap table** listing providers by average gap to AWS, with expandable rows showing per-category service alternatives (EU equivalent name, or “No alternative” for N verdicts).

The two visualisations are designed to serve complementary analytical purposes: the radar chart gives an immediate gestalt impression of a provider’s overall readiness profile relative to AWS, while the area chart and gap table support drill-down analysis at the dimension and service level. Clicking any data point in the area chart pins a detailed service table below the chart, listing AWS service name, coverage verdict, EU equivalent, and source reference for every service in that dimension.

*[SCREENSHOT REQUIRED: Cloud Readiness Assessment page showing the radar chart (left) and dimension breakdown area chart (right) with at least 4 providers visible, including AWS. Capture with the provider toggle buttons visible at the top.]*

Figure 5.6: Cloud Readiness Assessment: coverage profile radar (left) and dimension breakdown area chart (right).

*[SCREENSHOT REQUIRED: Gap summary table with one provider row expanded, showing the per-category service grid with EU alternative names and “No alternative” labels. Capture a provider that has a mix of Y, P, and N verdicts (e.g. Hetzner or Scaleway).]*

Figure 5.7: Gap summary table with an expanded provider row showing per-service EU alternatives.

#### 5.4.6 Preliminary Readiness Findings

The readiness data reveals a pronounced stratification among EU providers that correlates closely with the depth and maturity of their managed-service portfolios. T-Systems OTC achieves the highest individual dimension scores among all EU providers, recording a scalability meta-score of 72 and a performance meta-score of 87 out of 100 — reflecting its wide managed-service catalogue including FunctionGraph (serverless), ModelArts (ML platform), IoTDA, DMS Kafka, and CodeArts. T Cloud Public, assessed separately as a distinct commercial product from the same parent, achieves a marginally higher scalability score of 74 and the same performance score of 87, reflecting its purpose-built sovereign infrastructure design and planned Netherlands capacity expansion. OVHcloud follows as the next-ranked EU provider (scalability 68, performance 83), driven by its breadth of managed databases and broad regional availability. Both providers score significantly higher than the infrastructure-centric providers at the lower end of the spectrum.

Hetzner, despite being one of Europe’s most widely used cloud providers by customer count, records a scalability score of just 35 — reflecting its deliberate focus on virtual machines, block storage, and bare-metal servers without native managed services for serverless, workflow orchestration, or AI/ML. STACKIT, despite the substantial financial backing of the Schwarz Group, scores similarly low on scalability (32) owing to its 2021 launch date and the resulting managed-service portfolio still under active development. This pattern underlines the critical role of time-in-market: providers with deeper operational histories of cloud service delivery — particularly OVHcloud (cloud IaaS since 2012) and T-Systems OTC (since 2016) — have broader catalogues than newer entrants.

Across all providers, the dimensions with the largest average gaps to the AWS baseline are AI/ML (Amazon Bedrock has no equivalent among any EU provider), Database (no EU equivalent to DynamoDB, Amazon Aurora, or Amazon Timestream), and Dev/Deploy (Step Functions has no EU managed equivalent). These structural gaps are consistent across providers of all sizes and

ages, suggesting that they reflect the maturity of the underlying managed-service ecosystem rather than any individual provider’s product strategy. Full findings with ranked provider scores and per-dimension gap analysis are discussed in Chapter 8.

## 5.5 Financial Analysis Methodology

### 5.5.1 Rationale and Scope

The SovScore and Cloud Readiness Assessment instruments characterise *what* EU providers offer and *how* their jurisdictional posture compares to the AWS baseline. Both instruments, however, are silent on whether the capital underpinning EU providers is sufficient to close the observed readiness gap, and whether the *source* of that capital has its own sovereignty implications. A third complementary analysis — the Financial Consideration module — addresses these questions directly. It asks: (1) at current investment velocity, when will EU providers reach AWS parity? (2) Is the capital already deployed proportionate to the readiness achieved? and (3) does the ownership and origin of that capital itself affect a provider’s sovereignty score?

This section describes the Financial Consideration module of the CloudSov tool: how it was built, how its data were collected and classified, and how its five interactive visualisations should be read. The quantitative findings and their implications for the thesis research questions are discussed in Section 8.5.

### 5.5.2 Data Sources and Collection

Financial data were assembled from four categories of publicly available sources:

1. **Company filings and investor relations:** Annual reports, IPO prospectuses, and earnings releases from OVHcloud (Euronext Paris, since November 2021) and IONOS SE (Frankfurt Stock Exchange, since February 2023). AWS revenue figures are drawn from Amazon’s Form 10-K annual reports and converted to EUR at the 2024 annual average rate of \$1 = €0.920.
2. **Press releases and investor announcements:** Capital raises, PE/VC transactions, parent-company investment pledges, debt issuances, and government contract awards reported by each provider or their controlling group (Schwarz Group, Iliad Group, Deutsche Telekom AG).
3. **IPCEI-CIS documentation:** The European Commission’s December 2023 approval decision for the Important Project of Common European Interest on Next Generation Cloud Infrastructure and Services (IPCEI-CIS), including the official beneficiary list and published national funding allocations for Scaleway, OVHcloud, and Deutsche Telekom.
4. **EC sovereign cloud tender records:** The European Commission’s April 2026 award notices for the €180M Sovereign Cloud Tender (Lots 1–3), which publicly disclosed winning consortia and per-lot contract values.

Where a provider’s financial data are not publicly disclosed — most notably STACKIT (a wholly-owned Schwarz Group internal unit with no separate reporting obligation), Scaleway (Iliad subsidiary), and T-Cloud Public (Deutsche Telekom subsidiary) — revenue figures are derived from parent-group disclosures, analyst commentary, and cross-referencing with readiness capability

indicators. These estimates are labelled as such in the tool interface. All monetary amounts are denominated in EUR; USD figures are converted using the rate above, applied uniformly across all years.

### 5.5.3 Investment Milestone Classification

Each discrete capital event is recorded as an investment milestone with the following structured attributes: provider key, year, nominal EUR amount (where disclosed), event type, a short label, and a narrative description. Events with undisclosed amounts are retained in the milestone timeline for completeness but excluded from all quantitative aggregations. Event types are consolidated into six funding source categories for the capital structure analysis:

- **Capex** — direct infrastructure expenditure by the provider or its parent group (hardware procurement, data centre construction, regional expansion).
- **Debt** — bank loans, revolving credit facilities, high-yield bonds, and EIB green loans.
- **Market Capital** — IPO proceeds and private equity / venture capital injections.
- **Parent** — discretionary capital commitments from the controlling parent entity (Schwarz Group for STACKIT; Iliad for Scaleway) not separately categorised as Capex.
- **EU Programmes** — IPCEI-CIS grant approvals and EC sovereign cloud tender contract awards.
- **Other** — strategic acquisitions, infrastructure asset transactions, and government service contracts not classified under the above categories.

### 5.5.4 Financial Consideration Module: Tool Implementation and Visualisations

The Financial Consideration module is the third module of the CloudSov web application, accessible via the **Financial** tab in the top navigation bar. It presents five interactive charts, each addressing a distinct analytical question about the financial position and trajectory of EU sovereign cloud providers relative to AWS. Unlike the SovScore and Readiness modules, this module is read-only: there are no verdict inputs. All chart data are derived at backend startup from the structured milestone and revenue datasets described in Sections 5.5.2 and 5.5.3.

#### Readiness Gap Closure — EU Providers vs AWS

Figure 5.8 shows each EU provider’s historical readiness trajectory as a solid coloured line from its cloud launch year to the current score, followed by a dashed projection at the same velocity until the score reaches 100 (AWS parity). The underlying model is linear: for a provider with cloud launch year  $y_0$  and average readiness score  $s$  at Q2 2026, the annualised velocity and projected parity year are defined as:

$$v = \frac{s}{2026 - y_0} \quad [\text{pts/yr}] \quad (5.5)$$

$$t_{\text{parity}} \approx 2026 + \frac{100 - s}{v} \quad (5.6)$$

where  $v$  is the annualised readiness velocity and  $t_{\text{parity}}$  is the projected year at which the provider reaches score = 100. Estimated parity years derived from Equations 5.5–5.6 are annotated on each trajectory. Two vertical reference lines are overlaid: a grey dashed marker at 2026 labelled *Now*, indicating the analysis cut-off, and an amber dashed marker at 2030 labelled *EC Digital Decade*, corresponding to the headline cloud adoption target of the EU Digital Decade Policy Programme [11] (see Section 2.6). The chart addresses the question: *at current investment velocity, when will each EU provider reach AWS readiness parity, and which providers are on track to do so within the Digital Decade horizon?*

*[SCREENSHOT REQUIRED: Financial Consideration page — Readiness Gap Closure projection chart showing solid historical lines and dashed projected trajectories for all five EU providers, with AWS parity line at score = 100, annotated parity year labels, the grey “Now” marker at 2026, and the amber “EC Digital Decade” marker at 2030.]*

Figure 5.8: Readiness gap closure: historical readiness velocities (solid) projected forward (dashed) to AWS parity (score = 100). The amber vertical line marks the EC Digital Decade 2030 target [11].

### Annual Revenue Scale — EU Providers vs AWS

Figure 5.9 plots annual cloud-segment revenue on a log-scale Y-axis for AWS and the five EU providers across 2019–2024. A logarithmic scale is necessary because plotting AWS and the EU cohort on a linear axis would render the EU provider lines indistinguishable from zero: the multi-order-of-magnitude difference between AWS (€92B in 2024) and the highest-revenue EU provider (IONOS, €1.3B) is only legible on a log scale. The chart addresses the question: *how large is the revenue gap between AWS and EU providers, and is it narrowing or widening?*

### Investment and Funding Milestone Timeline

Figure 5.10 presents a full-width bubble chart that places each recorded investment milestone against calendar year on the X-axis. Bubble area is proportional to the committed amount (EUR); colour encodes the provider. Hovering over a bubble reveals the event label, year, amount, and event type. Milestones with undisclosed amounts are displayed at a fixed minimum size with a dashed outline, distinguishing them from confirmed capital events. AWS milestones are included for scale reference. The chart addresses the question: *what discrete capital events have shaped each provider’s trajectory, and how do they compare in scale to AWS’s infrastructure commitments?*

*[SCREENSHOT REQUIRED: Financial Consideration page — Revenue Scale Comparison (log Y-axis) showing AWS and the five EU providers as coloured lines, 2019–2024. The AWS line should be clearly visible at the top of the log scale; EU provider lines clustered at the lower end.]*

Figure 5.9: Annual cloud-segment revenue (EUR M, log scale): AWS vs the five EU providers, 2019–2024. A log scale is required to render both AWS and EU providers on a common axis.

*[SCREENSHOT REQUIRED: Financial Consideration page — Investment and Funding Milestone Timeline bubble chart. Capture with STACKIT's €11B Bitterfeld-Wolfen bubble clearly visible (the largest bubble in the chart) alongside the AWS reference bubbles. All provider colours should be visible in the legend. Dashed-outline bubbles indicate undisclosed amounts.]*

Figure 5.10: Investment and funding milestones for AWS and the five EU providers, 2006–2026. Bubble area is proportional to the committed capital (EUR); dashed outlines indicate undisclosed amounts.

### Investment Efficiency — Readiness Points per Euro Invested

Figure 5.11 is an investment efficiency scatter in which each EU provider is plotted with cumulative disclosed capital on the log-scale X-axis (€M) and current average readiness score on the Y-axis. The efficiency ratio  $\eta$  plotted beneath each provider label is defined as:

$$\eta = \frac{s}{C_{\text{disclosed}}} \quad [\text{pts}/\text{€B}] \quad (5.7)$$

where  $s$  is the provider’s average readiness score and  $C_{\text{disclosed}}$  is the cumulative disclosed capital in billions of euros. Only capital events with a publicly confirmed nominal amount contribute to  $C_{\text{disclosed}}$ ; milestones with undisclosed amounts are excluded from the denominator. Each data point displays the provider name above and  $\eta$  below. Providers positioned towards the upper-left — high readiness relative to low disclosed capital — represent the most capital-efficient paths to sovereignty. The chart addresses the question: *for every euro disclosed as invested, how much measurable cloud readiness has each provider achieved?*

*[SCREENSHOT REQUIRED: Financial Consideration page — Investment Efficiency scatter chart. Each EU provider plotted as a labelled dot with provider name above and pts/€B ratio below. Log X-axis (cumulative disclosed capital, €M), linear Y-axis (readiness score 0–100).]*

Figure 5.11: Investment efficiency scatter: current average readiness score vs cumulative disclosed capital (log scale, €M). The efficiency ratio  $\eta$  (readiness points per €B) is shown beneath each provider label.

### Capital Structure by Funding Source

Figure 5.12 is a stacked bar chart decomposing each provider’s total disclosed capital into the six funding source categories defined in Section 5.5.3. Each bar segment is colour-coded by category (Capex, Debt, Market Capital, Parent, EU Programmes, Other), and the chart is sorted to make the dominant funding model of each provider immediately legible. The chart addresses the question: *where does each provider’s capital come from, and does that origin carry its own sovereignty implications under the SovScore framework?*

### 5.5.5 Preliminary Financial Findings

The financial analysis surfaces three immediate observations that contextualise the SovScore and Readiness results established in preceding sections.

First, the revenue gap between AWS and the EU cohort is not merely large — it is structurally widening. AWS’s sustained 17% annual CAGR means the absolute revenue gap grows by €13–16B per year even as EU providers achieve their own growth. No EU provider in the study is on a trajectory to close this gap through organic revenue growth alone in a policy-relevant timeframe.

*[SCREENSHOT REQUIRED: Financial Consideration page — Capital Structure stacked bar chart. Each provider shown as a stacked bar with segments colour-coded by funding category (Capex, Debt, Market Capital, Parent, EU Programmes, Other). STACKIT's bar should dominate the chart due to the €11B parent commitment; OVHcloud and IONOS should show Market Capital segments.]*

Figure 5.12: Capital structure by funding source: total disclosed capital decomposed by category for each EU provider (€M). Undisclosed events are excluded.

Second, the gap-closure projection reveals a two-tier EU cohort. T-Systems (OTC / T-Cloud Public) and STACKIT are on trajectories that place AWS readiness parity within reach by approximately 2029–2031, driven by a combination of operational maturity (T-Systems) and accelerating capital deployment (STACKIT). OVHcloud, Scaleway, and IONOS, by contrast, face parity timelines extending into the late 2030s and 2040s under the current investment velocity.

Third, and most directly relevant to the thesis research question on sovereignty, the capital structure decomposition reveals a clean divide: providers financed through EU parent groups or EU institutional channels (STACKIT, Scaleway, T-Systems) achieve full Y verdicts on the SovScore questions that directly score capital origin (SOV-1 Q3, SOV-2 Q2). Providers with US private equity or public market capital exposure (OVHcloud via KKR and General Atlantic; IONOS via Warburg Pincus) receive P verdicts on these same questions. Capital structure is not merely a financial characteristic — it is a direct sovereignty determinant within the scoring framework used in this thesis. A full discussion of these findings, including the causal mechanism through which capital origin affects sovereignty scoring, is presented in Section 8.5.

### 5.5.6 Limitations

The financial analysis is subject to several important limitations. First, only *disclosed* capital events contribute to quantitative aggregations; providers with limited public reporting obligations (STACKIT, Scaleway, T-Cloud Public) are understated in the capital efficiency and capital structure analyses. Second, the revenue series for non-listed providers are management estimates subject to revision. Third, investment milestones represent *announced* or *committed* amounts, which may differ from capital actually deployed — most notably STACKIT's €11B Bitterfeld-Wolfen campus pledge, where Phase 1 construction is targeted for completion in 2027 and the associated managed services have not yet been delivered. Fourth, the uniform EUR/USD conversion rate does not account for multi-year exchange-rate fluctuation. These limitations are acknowledged in the discussion in Section 8.5.

## 5.6 Benchmark Pipeline Design

...

**5.6.1 Compute (sysbench)**

**5.6.2 Storage (fio)**

**5.6.3 Network (iperf3)**

**5.6.4 AI/ML Inference (Odin)**

**5.7 Cost Evaluation Methodology (TCO)**

...

**5.8 Deployment Experiment Design (Phase 3)**

...

**5.9 Evaluation Criteria and Metrics Summary**

...

## Chapter 6

# Technical Benchmarking and Price Consideration

In this chapter, we consider the technical benchmarking for all the providers, we would only consider our top 5 ranked providers from the readiness evaluation and measurements.

On the cloudSov, a web page was added to show the benchmarking results, and the data was compiled for each provider from multiple sources.

Gillam et al. in their paper carried out a The Fair Benchmarking project, was conducted at the University of Surrey between 1st April 2011 and 31st January 2012 to attempt to address this vexed question of value-for-money

Compute resource benchmarks are an established part of the high performance computing(HPC) research landscape

Gillam et al, also tried to address the problem of different machine comparability from different providers, where he stated it like "comparing apples to lizards"

Cloud Computing benchmarks need to offer up comparability for all parts of the IaaS, and should also be informative about all parts of the lifecycle of cloud system use. Individually, most existing benchmarks do not offer such comparability

Fortunately in the past, there have been some research papers that have addressed this issue, and how to accurately match the machines across providers, and fair comparison methodology.

Gray [14] identified four criteria for a successful benchmark: (i) relevance to an application domain; (ii) portability to allow benchmarking of different systems; (iii) scalability to support benchmarking large systems; and (iv) simplicity so the results are understandable. This drove my need to narrow down on this research in order to come up with a kind of benchmarking that is relevant to the providers selection and to accurately reflect the performance of the providers, and to be able to compare them in a fair manner. But unfortunately, this was not as easy as it seemed, especially since each providers have different machine specifications, and so to steer a fair comparison, I had to come up with a way to make the machines equivalent, and this was not an easy task.

Take Note that Benchmarks measures visualized machines and not the actual physical characteristics. In terms of scalability, we can break these into two types – horizontal scaling, involving the use of a number of machines running in parallel, and vertical scaling, involving different sizes of individual machines. In terms of the performance of each Cloud instance, vertical scaling is more

of interest here.

A series of reports on Cloud Performance Benchmarks [17] covers networking and CPU speed for AWS and Rackspace, as well as Simple Queue Service (SQS) and Elastic Load Balancing (ELB) in AWS

Jackson et al. [18] analysed HPC capabilities in AWS using NERSC – which needs MPI - to find that “the more communication, the worse the performance becomes”.

In the past, there were many websites including commercial web services that claimed to offer comparisons of cloud systems, but unfortunately, these are no longer available, such as CloudSleuth(CLOUDSLEUTH [21] offers visualisation of response times and availability for Cloud provider data centres, overlaid onto a World map.), CloudHarmony (CloudHarmony [20] has been monitoring the availability of Cloud services (38 in total) since late 2009.), and OpenBenchmarking.org (The OpenBenchmarking [22] website offers information from benchmark runs based around the Phoronix Test Suite. this is still active as of the time of this research ) , www.cloud-mercato.com and vpsbenchmark.com . but none offer a comprehensive comparison of all the providers .....

In this chapter, the cloudsof tool was used.

Another thing to consider was the “sense if the network I/O”, which region the test was run, a table of Cloud instance types used for benchmark tests for each provider respectively, and seen from our screenshot from cloudsof.

**BENCHMARKING SELECTION** Benchmark selection Cloud Computing benchmarks need to be able to account for the components of the Cloud “machine”. Typically, in IaaS, the components are the virtualized resources being provided to a virtual machine.

Traditional benchmarking involves the benchmark application being ready to run, and information captured about the system only from the benchmark itself.

For our purposes, we want to understand performance in terms of what you can get in a Cloud instance and require a set of benchmarks that tests different aspects of such an instance. Further, we also want to understand an underlying cost (in time and/or money) as necessary to achieve that performance. Put another way, if two Cloud instances achieved equal performance in a benchmark, but one took equally as long to be provisioned as it did to run the benchmark, what might we say about the value-for-money question? Raw performance is just one part of this. Additionally, if performance is variable for Cloud instances of the same type from the same provider, would we want to report only “best” performance? We can also enquire as to whether performance varies over time – either for new instances of the same type, or in relation to continuous use of the same instances over a period

We need to consider the 3 Sysbench, Geekbench, Phoronix test suite and why we opted for the Phoronix test suite, and how it is more relevant to our research.

The selected benchmarks are intended to treat (i) Memory IO; (ii) CPU; (iii) Disk IO; (iv) Application; (v) Network

1. Memory IO (short description of the benchmark and why it is important) 2. STREAM 3. CPU (short description of the benchmark and why it is important) 4. LINPACK 5. Disk I/O (short description of the benchmark and why it is important) 6. Bonnie++ 7. IOzone 8. Bzip2 9. Network (short description of the benchmark and why it is important) 10. Iperf 11. MPPTEST 12. Simple speed tests 13.

All but one of the benchmarks was run across all selected Cloud providers within selected regions on all available machine types. 10 instances of each type were run to obtain information about variability. Due to the sheer number of results obtained, in this section we limit our focus primarily to one set of (relatively) comparable machine types across the four providers

Also Kris Lowet, In his research, "benchmark between cloud servers" (January 2024) where him and his team compared the performance of cloud servers from Hetzner, digitalocean, Linode, Vultr, OVH, Upcloud and scaleway. (some of our top ranked readiness providers).....

they only compared the technical performance of equivalent virtual machines and did not take into account other services offered by the providers, the level of support, SLAs, audits, reliability, global presence, etc.

They classified the benchmarks differently, with the focus on the system, a single and multi-core CPU, the RAM and the storage space.

1. pts/apache (focus on the system). This is a test of the Apache HTTPD web server. This Apache HTTPD web server benchmark test profile makes use of the Golang "Bombardier" program for facilitating the HTTP requests over a fixed period time with a configurable number of concurrent clients. (briefly discuss their results for this benchmark.....) 2. pts/hint (focus on single core CPU). This test runs the U.S. Department of Energy's Ames Laboratory Hierarchical INTeграtion (HINT) benchmark. (briefly discuss their results for this benchmark.....) 3. pts/compress-7zip (focus on multi core CPU). This is a test of 7-Zip compression/decompression with its integrated benchmark feature. (briefly discuss their results for this benchmark.....) 4. pts/stream (focus on the memory). This is a benchmark of Stream, the popular system memory (RAM) benchmark. (briefly discuss their results for this benchmark.....) 5. pts/postmark (focus on the storage). This is a test of NetApp's PostMark benchmark designed to simulate small-file testing similar to the tasks endured by web and mail servers. This test profile will set PostMark to perform 25,000 transactions with 500 files simultaneously with the file sizes ranging between 5 and 512 kilobytes. (briefly discuss their results for this benchmark.....)

Considering the above, I have decided to adopt some of these benchmarks, (explain how we matched both benchmarks that are similar) and also add some more benchmarks that are relevant to our research. - Performance variation is a question of Quality of Service (QoS), and service level agreements (SLAs) tend only to offer compensation when entire services have outages, not when performance dips - booting time - setup time - - - - - compute costs (justify why these benchmarks are important and relevant to our research, and how they will help us in the provider selection process.....)

To better suit the benchmarks below is the table of instances from each provider, what VM was selected for the providers with similar specifications, and the region where the benchmark was run.

(then explain for each benchmark how the data was collected for each instances of the various providers with sources ..... ) Here the data was collected and saved as json files in our cloudsof tool backend, (explain briefly.....) I like how Gillam et al, did 10 runs, took max, min, average for each benchmark and showed the results, (check the table), he said We store benchmark data in a directory structure based on benchmark, provider, operating system, region, date, and machine size (We should state our collected data was stored in our tool)

(on the frontend, now explain all the visualizations of the data ..... and comparison charts..... )

Each visualization and table must be explained like what they did "Table 11 A table, as generated by the software, showing results for both STREAM and Bonnie++ on AWS"

Lastly what inferences can be drawn from the data, that relation to sovereignty

## Chapter 7

# Deployment Experiment

### 7.1 Selected Provider and Rationale

...

### 7.2 EU Stack Setup (Mosquitto/EMQX, FastAPI, Grafana, Prometheus)

...

### 7.3 AWS Stack Setup (IoT Core, SageMaker, CloudWatch)

...

### 7.4 Workload Simulation and Results

...

### 7.5 Latency Analysis (p50, p95, p99)

...

### 7.6 Operational Friction Log

...

## 7.7 Cost Comparison

...

# Chapter 8

## Discussion

This chapter synthesises the findings from the two experiments — the Sovereignty Scoring Framework (SovScore) and the Cloud Readiness Assessment — and situates them within the broader context of EU cloud policy, enterprise procurement, and the evolving geopolitics of cloud infrastructure. Section 8.1 answers the three research questions directly. Section 8.2 develops the sovereignty-readiness trade-off as a structural insight. Section 8.3 examines the Huawei dependency in T-Systems as the most nuanced finding of the study. Section 8.4 analyses the role of cloud maturity and capital investment in explaining the readiness gap. Section 8.6 discusses the SOV-5 and SOV-6 structural constraints that represent a practical ceiling for all providers. Section 8.7 addresses vendor lock-in implications. Section 8.8 offers actionable recommendations for EU enterprises. Section 8.9 closes with a candid account of the study’s limitations.

### 8.1 Research Question Responses

#### 8.1.1 RQ1: Can EU cloud providers satisfy the EC DG DIGIT Cloud Sovereignty Framework requirements?

The short answer is: partially, and in a stratified way. All 18 evaluated providers satisfy at least some of the eight sovereignty objectives defined in the EC DG DIGIT Cloud Sovereignty Framework v1.2.1 [13]. However, no provider achieves a SEAL-4 maturity level across all eight objectives simultaneously. The most consistently satisfied objectives are SOV-3 (data residency and GDPR alignment), SOV-7 (incident response and audit), and SOV-8 (contractual transparency), reflecting the comparatively mature state of EU data-protection compliance following the GDPR [14] and the Schrems II ruling [6], which created strong regulatory incentives for providers to formalise these commitments. BSI C5, ISO 27001, and SOC 2 certifications — held by most providers in the study — directly support these objectives.

The most persistently failed or partially satisfied objectives are SOV-5 (hardware and firmware sovereignty) and SOV-6 (freedom from non-EU high-performance computing dependencies). These are not failures of governance or legal structure; they are structural constraints imposed by the global hardware supply chain (see Section 8.6). Even the highest-performing providers cannot satisfy SOV-5 Q2 (firmware of EU origin) or SOV-6 Q4 (HPC silicon from EU/trusted sources) because no commercially viable European alternative to Intel, AMD, or NVIDIA processors and

their associated firmware exists at scale.

SOV-2 (freedom from non-EU extraterritorial law) presents a more nuanced picture. Providers with a clean EU-only corporate structure — such as Infomaniak (Swiss law), Cleura (Swedish law), and Elastx (Swedish law) — score fully on this objective. Providers with partial dependencies on non-EU parent companies, hardware vendors, or technology suppliers receive partial verdicts. T-Systems OTC receives multiple partial verdicts under SOV-2 due to its Huawei technology dependency, discussed at length in Section 8.3.

The conclusion for RQ1 is that the EC DG DIGIT framework identifies real and meaningful distinctions between EU cloud providers, but it also exposes two categories of constraint: those that are addressable through governance and investment decisions (corporate structure, supply-chain diversification), and those that require systemic, industry-level or policy-level solutions (EU semiconductor independence, open-source firmware ecosystems).

Table 8.1 summarises the per-objective SEAL levels and aggregate SovScore for the five EU providers forming the financial analysis cohort.

Table 8.1: SovScore results for the five EU providers under financial analysis: per-objective SEAL levels (0–4), weighted aggregate score (0–100), and pass/fail determination against minimum SEAL thresholds. Superscript † denotes a cell below the minimum threshold for that objective. Minimum SEAL thresholds per objective: SOV-1 $\geq$ 3, SOV-2 $\geq$ 4, SOV-3 $\geq$ 3, SOV-4 $\geq$ 2, SOV-5 $\geq$ 2, SOV-6 $\geq$ 2, SOV-7 $\geq$ 3, SOV-8 $\geq$ 1. A provider passes only if all eight thresholds are met simultaneously. T-Systems = T-Systems (OTC / T-Cloud Public), merged readiness entity.

| Provider  | SOV-1<br>(15%) | SOV-2<br>(10%) | SOV-3<br>(10%) | SOV-4<br>(15%) | SOV-5<br>(20%) | SOV-6<br>(15%) | SOV-7<br>(10%) | SOV-8<br>(5%) | Score | Pass |
|-----------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|---------------|-------|------|
| STACKIT   | 4              | 4              | 3              | 4              | 2              | 3              | 4              | 3             | 82.5  | ✓    |
| Scaleway  | 4              | 3 <sup>†</sup> | 4              | 4              | 1 <sup>†</sup> | 2              | 3              | 4             | 72.5  | ×    |
| T-Systems | 4              | 2 <sup>†</sup> | 3              | 3              | 1 <sup>†</sup> | 2              | 4              | 4             | 66.3  | ×    |
| OVHcloud  | 3              | 1 <sup>†</sup> | 2 <sup>†</sup> | 3              | 2              | 2              | 3              | 3             | 58.8  | ×    |
| IONOS     | 2 <sup>†</sup> | 2 <sup>†</sup> | 2 <sup>†</sup> | 3              | 1 <sup>†</sup> | 2              | 3              | 3             | 52.5  | ×    |
| Min. SEAL | $\geq 3$       | $\geq 4$       | $\geq 3$       | $\geq 2$       | $\geq 2$       | $\geq 2$       | $\geq 3$       | $\geq 1$      |       |      |

### 8.1.2 RQ2: How does the readiness of EU providers compare to AWS?

A significant, stratified gap exists between EU providers and AWS across all 13 dimensions of the Cloud Readiness Assessment. AWS, used as the 100% baseline, holds advantages that span three distinct categories.

**Universal gaps** — services for which no EU provider offers even a partial equivalent — include Amazon Bedrock (managed generative-AI API), Amazon Aurora (fully managed relational database with multi-master replication), AWS Step Functions (managed workflow orchestration), and several specialised ML services. These represent categories in which AWS has made sustained platform-level investments over multiple years and where the service is architecturally integrated into the broader AWS ecosystem in ways that cannot be replicated by simply deploying an open-source alternative.

**Near-universal gaps** — services for which only one or two providers offer partial equivalents — include Amazon DynamoDB (serverless NoSQL at scale), Amazon Cognito (customer identity and access management), and AWS IAM Identity Center (enterprise single sign-on). The handful

of providers offering partial equivalents typically do so through managed deployments of open-source products (e.g., Keycloak for CIAM) rather than purpose-built managed services, meaning operational maturity and SLA parity remain open questions.

**Stratified gaps** — services where the best EU providers approach AWS coverage but the field is wide — are most visible in core IaaS (compute, object storage, VPC networking) and basic PaaS (managed Kubernetes, managed PostgreSQL). Here, T-Systems OTC, OVHcloud, and IONOS come within 10–20 percentage points of AWS.

At the provider level, T-Systems (OTC / T-Cloud Public) achieves the highest EU readiness scores in this cohort: scalability 74/100 and performance 87/100, which makes it the closest EU analogue to AWS in terms of managed-service breadth. OVHcloud follows (scalability 68, performance 83). At the other end, STACKIT (scalability 32, performance 80) and Hetzner (scalability 35, performance 72) reflect younger or more IaaS-focused portfolios.

The conclusion for RQ2 is that EU providers are not uniformly lagging. The best EU providers cover the overwhelming majority of commodity cloud needs for a typical enterprise. The gap becomes decisive precisely in the areas of AI/ML, serverless, and native orchestration — services that represent the fastest-growing share of enterprise cloud spending [17].

Table 8.2 presents the full per-dimension scores for the five EU providers, ordered by average readiness score. Colour bands follow the convention used in the CloudSov readiness module: green (score >60), yellow (41–60), orange (21–40), red ( $\leq 20$ ).

Table 8.2: Cloud Readiness Assessment dimension scores (0–100) for the five EU providers. Providers are ordered by average readiness score (highest to lowest). Scalability and Performance are pre-scored meta-dimensions; all other dimensions are computed from service-level coverage verdicts. T-Systems = T-Systems (OTC / T-Cloud Public) merged entity, using T-Cloud Public assessment data.

| Dimension      | T-Systems   | OVHcloud    | STACKIT     | Scaleway    | IONOS       |
|----------------|-------------|-------------|-------------|-------------|-------------|
| Scalability    | 74          | 68          | 32          | 42          | 45          |
| Performance    | 87          | 83          | 80          | 75          | 78          |
| Application    | 50          | 50          | 0           | 0           | 0           |
| Compute        | 75          | 67          | 67          | 83          | 33          |
| Storage        | 90          | 90          | 70          | 60          | 50          |
| Database       | 43          | 43          | 36          | 29          | 14          |
| Networking     | 93          | 93          | 79          | 50          | 50          |
| AI/ML          | 50          | 17          | 8           | 8           | 0           |
| Security       | 67          | 62          | 79          | 42          | 21          |
| Monitoring     | 88          | 62          | 62          | 50          | 38          |
| Dev/Deploy     | 70          | 30          | 30          | 20          | 20          |
| IoT            | 100         | 25          | 0           | 25          | 0           |
| Messaging      | 50          | 40          | 30          | 40          | 0           |
| <b>Average</b> | <b>72.1</b> | <b>56.2</b> | <b>44.1</b> | <b>40.3</b> | <b>26.8</b> |

### 8.1.3 RQ3: What is the trade-off between sovereignty and readiness?

The trade-off is real but not absolute. Contrary to the naive expectation that the most sovereign provider would simply be the least capable, the data reveal a more structured relationship: providers

that maximise sovereignty through corporate simplicity, EU-only supply chains, and avoidance of non-EU technology tend to score well on most sovereignty objectives and poorly on readiness dimensions that require sustained capital investment in proprietary managed services. Conversely, providers that have built the broadest managed-service portfolios — T-Systems OTC, OVHcloud — have done so partly by incorporating non-EU components (Huawei infrastructure in OTC’s case, US hyperscaler-origin software stacks in others) that introduce sovereignty compromises.

The trade-off is best understood as a *frontier*, not a binary. Section 8.2 develops this further. The practical implication for RQ3 is that no single EU provider currently sits at the Pareto frontier of both maximum sovereignty and maximum readiness, and that the policy objective of a “European cloud” that equals AWS on capability while exceeding it on sovereignty is an aspiration requiring both sustained investment and structural supply-chain change that is not achievable within the horizon of this study.

## 8.2 The Sovereignty–Readiness Trade-off

The most durable conceptual contribution of this study is the articulation of what might be termed the *sovereignty–readiness frontier* — an implicit Pareto curve on which each provider occupies a position that reflects the choices it has made in balancing regulatory compliance, corporate structure, and managed-service investment.

Figure 8.1 illustrates this frontier conceptually. At one extreme sits locally hosted infrastructure: maximum sovereignty (no external dependencies, no extraterritorial exposure, no non-EU hardware procurement beyond the structural HW constraints discussed in Section 8.6) and minimum readiness (no managed databases, no AI/ML APIs, no serverless, no CDN). At the other extreme sits AWS: maximum readiness (the reference baseline of 100% for every dimension) and minimum sovereignty from the EU perspective, given its exposure to the US CLOUD Act [6], the absence of EU-headquartered governance, and the full weight of US export-control jurisdiction.

Several observations follow from locating providers on this frontier.

**Smaller, purer EU providers cluster toward the sovereignty pole.** Infomaniak, Elastx, and Cleura have simple EU-only corporate structures, no Chinese or US parent entities, and operate primarily on open-source IaaS stacks (OpenStack). Their SovScore verdicts are consistently clean: no partial or negative verdicts under SOV-2 or SOV-4. Yet their readiness scores reflect the bounded managed-service portfolios of providers that have not pursued the capital-intensive path of building proprietary PaaS layers.

**Larger, more-certified providers cluster toward the readiness pole — but with sovereignty costs.** T-Systems OTC has the broadest managed-service catalogue among EU providers studied. It achieves BSI C5 Type 2 attestation and operates under German law. Yet its Huawei infrastructure dependency introduces the SOV-2 and SOV-5 partial verdicts that push it away from the sovereignty pole. This is the central paradox: the provider most often cited as a model of EU sovereign cloud is also the provider with the most significant non-EU supply-chain exposure.

**The frontier is dynamic.** Providers can move along it. Deutsche Telekom’s investment in T Cloud Public — a sovereignty-first successor to OTC with Huawei access to the production environment contractually prohibited — represents a deliberate strategic shift toward the sovereignty pole without sacrificing the managed-service portfolio that defines T-Systems’ readiness position. T Cloud Public still runs on Huawei FusionSphere as its core cloud OS, but the governance controls formalised around that dependency already produce a materially higher sovereignty score. If T

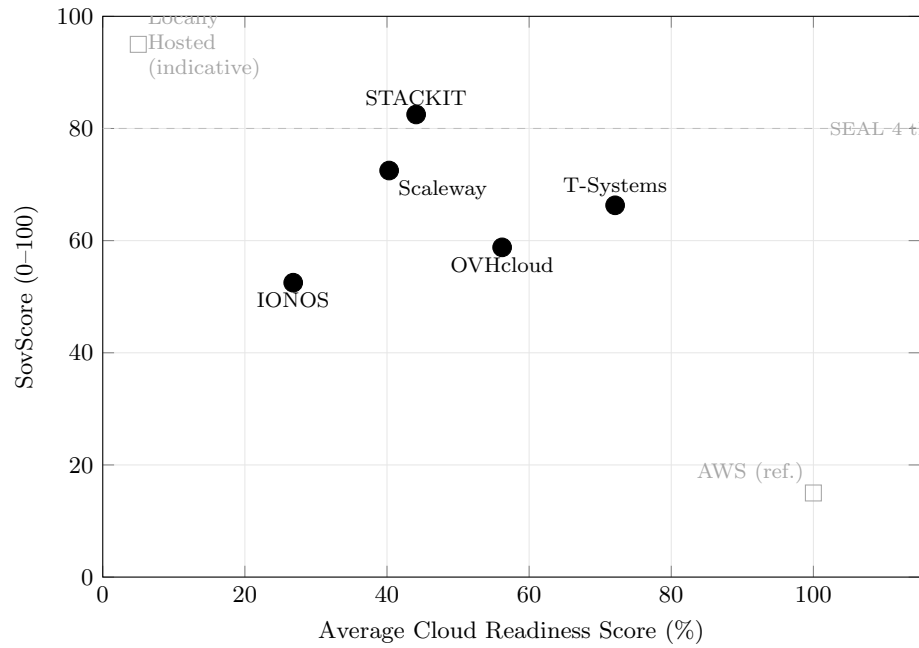


Figure 8.1: Sovereignty-readiness plane for the five EU providers under financial analysis, plotted using actual computed SovScore and average cloud readiness scores. AWS (readiness baseline = 100%) and a locally hosted boundary marker are shown for reference. The dashed line marks the SEAL 4 SovScore threshold (80%). No provider currently occupies the upper-right quadrant (high sovereignty *and* high readiness).

Cloud Public fully supplants OTC and completes its supply-chain diversification away from Huawei, T-Systems would move to a more Pareto-optimal location on the frontier. Section 8.3 examines this trajectory in depth.

**The AI/ML gap creates a new dimension of the frontier.** The universal absence of a Bedrock-equivalent across all EU providers means that the frontier has an additional dimension that cannot currently be addressed by any EU provider at all. This is not simply a matter of investment or time: building a frontier AI API requires access to large-scale GPU clusters (constrained by SOV-6), proprietary foundation model IP (where EU providers have no commercially deployed equivalent), and an inference infrastructure at global scale. The implication is that the sovereignty-readiness frontier as described here understates the total gap, because the AI/ML dimension is currently truncated for all EU providers.

## 8.3 The Huawei Problem and Deutsche Telekom’s Strategic Response

The most nuanced finding of the SovScore assessment concerns T-Systems. T Cloud Public — the platform Deutsche Telekom is developing as the successor to the Open Telekom Cloud (OTC) — holds BSI C5 Type 2 attestation, is approved for German federal administration under the GovTech framework, and operates under the full jurisdiction of German and EU law. By conventional measures it is one of the most compliance-certified cloud platforms in Europe. And yet the SovScore assessment assigns it multiple partial and negative verdicts under SOV-2 and SOV-5 because of its historical and partially ongoing dependency on Huawei FusionSphere infrastructure.

This finding deserves careful unpacking, because it illustrates both the analytical power and the interpretive limits of a framework-based scoring approach.

### 8.3.1 The Legal Exposure Under SOV-2

SOV-2 assesses freedom from non-EU extraterritorial law. The partial verdict for SOV-2 Q2 (*free from non-EU extraterritorial law*) arises from China’s National Intelligence Law of 2017 (NIS), which obliges Chinese organisations and citizens to “support, assist, and cooperate with state intelligence work.” Huawei, as a Chinese-headquartered corporation with deep operational ties to the Chinese state, is subject to this law regardless of where its products are deployed.<sup>1</sup>

The distinction that T-Systems draws — and that the GovTech approval implicitly accepts — is between *operational* access and *legal-structural* exposure. T-Systems maintains that Huawei has no runtime access to customer data and that network segmentation, access controls, and audit mechanisms prevent any such access in practice. This is a defensible technical claim. The sovereignty framework, however, assesses structural exposure rather than current operational practice, because the relevant question for a ten-year infrastructure contract is not whether Huawei *currently* has access but whether the legal architecture *creates conditions under which* it could be compelled to seek access. Under this reading, the partial verdict is analytically correct.

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<sup>1</sup>The Chinese NIS does not contain a territorial carve-out analogous to, for example, the GDPR’s adequacy mechanism. It applies to Chinese entities globally. Whether and how it can be enforced against a hardware vendor operating infrastructure on behalf of a German cloud provider has not been litigated, but the legal exposure is structural rather than theoretical.

SOV-2 Q3 (*no technical channels for non-EU data access*) receives a partial verdict because Huawei’s third-level hardware support model creates a technically documented channel — Huawei engineers providing firmware-level support remotely — that constitutes a potential access vector, irrespective of contractual prohibitions. Again, the distinction is between contractual access control (which exists) and architectural access control (which does not fully exist while Huawei engineers can reach the hardware layer).

### 8.3.2 The Hardware and Firmware Exposure Under SOV-5

SOV-5 Q2 (*firmware of EU origin*) receives a negative (N) verdict — the strongest negative available — because T Cloud Public servers run Huawei Baseboard Management Controller (BMC) firmware, Intel Management Engine (ME) firmware, and NVIDIA GPU firmware. None of these originate from EU entities. This is not unique to T-Systems: the same structural issue applies to every provider in the study (see Section 8.6). What distinguishes T-Systems is the additional Huawei BMC layer on top of the already-universal Intel ME and NVIDIA dependencies.

SOV-5 Q3 (*software stack of EU origin*) receives a partial verdict because FusionSphere — the virtualisation and orchestration layer co-developed by Huawei and Deutsche Telekom since 2016 — is substantially Huawei-developed. The co-development history is important: OTC was built from the start as a joint project, meaning T-Systems does not own the full IP stack independently of Huawei.

### 8.3.3 Deutsche Telekom’s Strategic Response

The significance of the Huawei findings is amplified by Deutsche Telekom’s response to them, which the SovScore assessment captures as a transitional rather than a permanent state.

As confirmed in personal communication with Prof. Dr. Oliver Kramer (TH Rosenheim, June 2026) [23], Deutsche Telekom has made a deliberate decision not to restructure OTC but instead to invest in an entirely new sovereign cloud infrastructure. T Cloud Public was launched as a direct response to the Huawei issue. T-Systems operates the platform exclusively with EU-resident staff and enforces a zero-access policy for Huawei personnel in production. However, both OTC and T Cloud Public continue to run on Huawei FusionSphere as the core cloud OS — a dependency T-Systems acknowledges on its own platform blog — meaning the Chinese technology dependency is managed through operational controls rather than eliminated at the architectural level. It has achieved BSI C5 Type 2 attestation and has received federal approval for use by German public administration under the GovTech framework. The spring 2026 milestone — incorporation of T Cloud Public into the GovTech framework agreement for cloud and AI services, enabling federal, state, and local authorities to procure it without individual tenders — represents a formal state certification that the German government considers the sovereignty risk to be managed.

Deutsche Telekom’s Industrial AI Cloud goes further: it explicitly excludes both Huawei technology *and* US hyperscaler software stacks, positioning itself as the most structurally clean enterprise AI offering in the German market. T-Systems has also initiated a dual-vendor hardware strategy to reduce Huawei hardware dependence across its remaining OTC deployments.

### 8.3.4 What the SovScore Assessment Captured

The SovScore results for T-Systems OTC reflect the platform as it existed during Q1–Q2 2026, at a moment of active transition. The multiple partial and negative verdicts under SOV-2 and

SOV-5 are accurate for OTC as deployed. They are, however, best understood as a snapshot of a transitional state rather than a permanent structural deficiency. The investment in T Cloud Public, the GovTech approval, and the Industrial AI Cloud exclusion policy collectively suggest that Deutsche Telekom has identified the same vulnerabilities that the SovScore framework surfaces and is actively remediating them.

This has two implications for interpreting the results. First, the SovScore framework is doing its job: it identified a real and substantive sovereignty vulnerability in the market leader, one serious enough that it prompted a strategic platform rebuild. Second, any comparative ranking that places T-Systems below smaller providers with simpler supply chains should be understood in its temporal context. A re-assessment of T Cloud Public in 2027 — once the OTC-to-TCP migration is complete and the dual-vendor hardware strategy is operational — may well produce materially different verdicts under SOV-2 and SOV-5.

### 8.3.5 The OTC–T Cloud Public Intra-Provider Comparison

A methodological decision of this study was to assess OTC and T Cloud Public as two separate provider entries rather than collapsing them into a single Deutsche Telekom entry. This choice is analytically justified by the fact that the two platforms differ in material ways on nearly every sovereignty dimension, despite sharing the same corporate parent, the same physical data centres, the same environmental practices, and the same primary service catalogue. The comparison between the two produces one of the study’s most concrete findings.

On the **SovScore** assessment, T Cloud Public achieves notably better results than OTC on SOV-1, SOV-2, SOV-4, and SOV-5 — the four objectives most directly affected by the Huawei dependency. Both OTC and T Cloud Public receive a Partial (P) verdict on SOV-2 Q2 (freedom from Chinese extraterritorial law) — neither can eliminate the structural legal risk arising from Huawei FusionSphere’s Chinese origin and Huawei’s obligations under China’s National Intelligence Law. The distinction between them falls on SOV-2 Q3 (no legal or technical channels for non-EU data access): T Cloud Public receives Partial (P) because Huawei’s engagement is limited to zero-access video consultation only, while OTC receives Negative (N) because Huawei engineers have participated more directly in platform lifecycle management without the same formally enforced zero-access constraint. On SOV-5 (Supply Chain), T Cloud Public has initiated a dual-vendor hardware strategy and is actively reducing FusionSphere dependence; OTC has no documented equivalent programme, resulting in two N verdicts (Q3 and Q4) where T Cloud Public receives Partial.

On the **Readiness Assessment**, the two platforms are closely matched because they share the same managed-service catalogue — ModelArts, FunctionGraph, CCE, DMS Kafka, and the rest of the OTC service portfolio are present in both. T Cloud Public receives a marginally higher scalability meta-score (74 vs. 72) reflecting the newer, purpose-built infrastructure design and its planned capacity expansion in the Netherlands. Performance meta-scores are nearly identical (87 vs. 87) given the shared Biere and Magdeburg data centre infrastructure. The readiness gap between the two platforms is therefore negligible — they serve the same enterprise workloads with the same service depth.

The strategic implication is significant: Deutsche Telekom has effectively split its cloud offering along a *sovereignty axis*. OTC continues to serve existing customers on a platform with a known Huawei dependency that makes it unsuitable for the most sovereignty-sensitive workloads. T Cloud Public targets new customers — particularly in the German public sector — who require a platform

that can pass the legal-control criteria under SOV-2. The spring 2026 GovTech framework approval of T Cloud Public is the German state’s formal endorsement of this distinction [23]. Enterprises still operating on OTC should treat migration to T Cloud Public not as an infrastructure upgrade but as a sovereignty compliance step.

## 8.4 Timeline and Investment: AWS versus EU Providers

The readiness gap between AWS and EU providers cannot be understood without accounting for the asymmetry in investment horizon and capital deployed. AWS launched its first commercial IaaS services in 2006, giving it a twenty-year runway of managed-service development by the time of this assessment. Its annual capital expenditure reached approximately \$53 billion USD in 2023 — a figure that exceeds the total capital raised by every European cloud provider combined, including over their full operating histories.

### 8.4.1 The Cloud Maturity Trajectory

A provider’s cloud operational age is a meaningful predictor of its readiness score. The data support this at a broad level: OVHcloud, which has operated IaaS since approximately 2012 (fourteen years of cloud experience), achieves a scalability score of 68; STACKIT, which launched as a cloud provider in 2021 (five years of cloud experience), achieves a scalability score of 32. The gap of 36 percentage points corresponds to approximately nine years of additional managed-service development. Hetzner (fourteen years) achieves a scalability score of 35, which appears to contradict this trajectory — but Hetzner’s cloud strategy has been explicitly IaaS-focused rather than PaaS-broadening, illustrating that time in the market is a necessary but not sufficient condition for readiness breadth.

OVHcloud’s cumulative investment of approximately €3 billion over its lifetime, including the 2021 IPO that raised €400 million [27], has produced a scalability score of 68 and performance score of 83. Extrapolating from the investment-to-score ratio implied by this data point and AWS’s trajectory is speculative, but it suggests that the gap is closable over a ten-to-fifteen-year horizon at current EU investment rates — and potentially faster if EU institutional co-financing accelerates the trajectory.

### 8.4.2 IPCEI-CIS and the European Cloud Strategy

The EU is not a passive observer of this investment gap. The Important Projects of Common European Interest – Cloud Infrastructure and Services (IPCEI-CIS) [12] provides EU co-financing for European cloud infrastructure, with T-Systems, OVHcloud, and a number of other providers participating. The European Cloud Strategy (2021–2030) [10] targets European cloud providers serving 40% of EU public sector workloads by 2030. These commitments create a procurement-driven demand pull that can justify the managed-service investment that EU providers currently lack the purely commercial incentive to fund.

STACKIT is the provider where this dynamic is most visible. Backed by Schwarz Group — the parent of Lidl and Kaufland, with revenues exceeding €113 billion in 2023 — STACKIT has announced cloud investments of over €300 million [30]. At five years of cloud operation, its current readiness scores (scalability 32, performance 80) reflect an early-stage managed-service portfolio. The Schwarz Group’s captive demand from its own retail and logistics operations, combined with

its announced investment trajectory, makes STACKIT the provider most likely to show the steepest readiness improvement curve over the 2026–2030 period. A re-assessment in 2028 is warranted.

Deutsche Telekom, with 2024 group revenue of €112 billion [8] and active IPCEI-CIS participation, occupies a different position: it has the financial scale of a hyperscaler parent but has historically under-invested in cloud managed services relative to its balance sheet. The T Cloud Public rebuild and the Industrial AI Cloud suggest that Deutsche Telekom is now deploying that balance sheet more aggressively toward cloud sovereignty.

### 8.4.3 The AI/ML Horizon

The GenAI service gap (Amazon Bedrock: universal absence) deserves particular attention in the investment context. AWS launched Bedrock in April 2023 following years of SageMaker investment. It therefore has a two-to-three-year head start in production-grade managed AI API infrastructure. EU providers that have announced AI cloud services — including T-Systems’ Industrial AI Cloud and OVHcloud’s AI Deploy platform — are in early or mid-stage deployment as of Q2 2026. Closing the GenAI gap requires not just investment in inference infrastructure but access to competitive foundation models (where the EU has emerging assets in Mistral, Aleph Alpha, and others) and a customer ecosystem willing to pay for EU-hosted AI inference.

The Digital Compass 2030 target, combined with the EU AI Act’s preference for high-risk AI systems deployed on auditable, sovereign infrastructure [15], creates a plausible procurement pathway for EU AI cloud services that could accelerate this investment. By 2030, the top EU providers are likely to offer competitive managed AI inference, though whether they will match the breadth of Bedrock’s model catalogue is uncertain.

## 8.5 Financial Landscape and Capital Efficiency of EU Sovereign Cloud Providers

The preceding section introduced investment trajectories at a narrative level, noting that STACKIT’s Schwarz Group backing and OVHcloud’s 2021 IPO represent qualitatively different capital strategies. This section deepens that analysis using the quantitative data collected in the CloudSov Financial Consideration module. Five visualisations are presented and interpreted in turn: a revenue scale comparison, a readiness gap-closure projection, an investment milestone timeline, a capital efficiency scatter, and a capital structure decomposition. Together they frame a central financial question of this thesis: *whether the existing capital base is sufficient to close the readiness gap with AWS, and whether the source of that capital makes EU providers more or less sovereign in the process.*

### 8.5.1 The Revenue Chasm: Scale Asymmetry Between AWS and EU Providers

The most immediate finding of the financial analysis is the scale asymmetry between AWS and every EU provider in the study. AWS generated approximately €92B in cloud revenue in 2024 (converted from USD 100.1B at the 2024 annual average rate)—a figure that exceeds the combined estimated revenues of all five EU providers by a factor exceeding fifty. IONOS, the largest EU provider by disclosed cloud revenue at approximately €1.4B for 2024, represents roughly 1.5% of AWS’s scale.

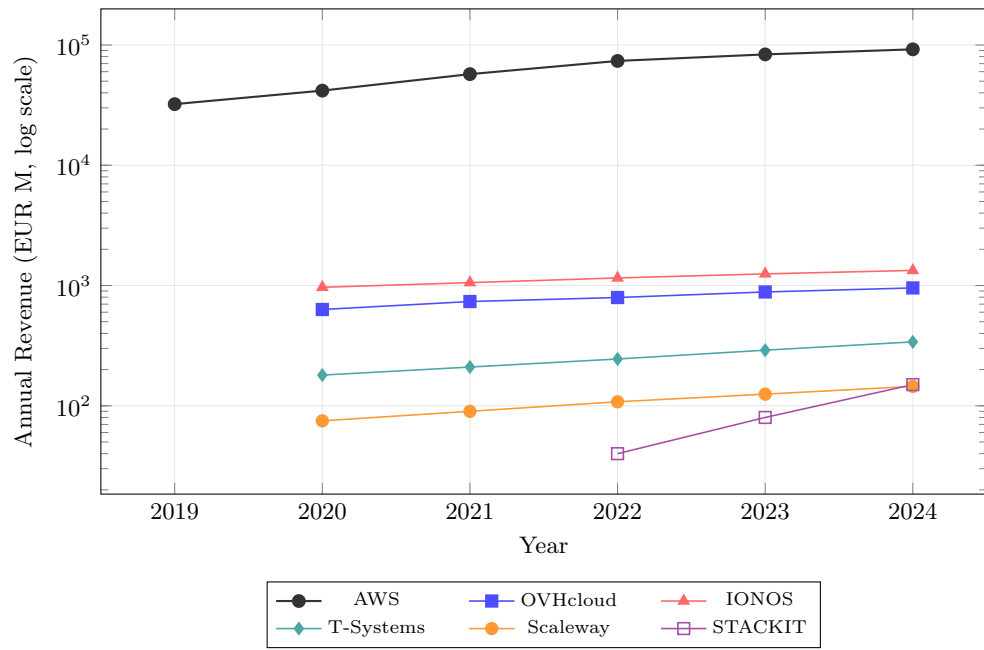


Figure 8.2: Annual cloud-segment revenue (EUR M, log scale) for AWS and the five EU providers, 2019–2024. A log scale is required to render both AWS and the EU providers on a common axis; on a linear scale EU provider lines would be indistinguishable from zero. STACKIT data begins at 2022 (launch 2021, revenue not publicly disclosed in first year).

The contrast is starker still for the smaller providers: STACKIT and Scaleway operate at revenues estimated between €200M and €400M, implying a gap of two orders of magnitude.

The revenue comparison is presented on a logarithmic scale (Figure 8.2) because a linear scale renders the EU provider lines effectively indistinguishable from zero. The log-scale representation, while compressing the gap visually, reveals the structural character of the asymmetry: AWS has sustained annual revenue growth of approximately 17% CAGR over the 2019–2024 period, meaning the absolute gap between AWS and the EU cohort *widens by approximately €13–16B per year* even as EU providers grow their own revenues.

This scale differential has direct implications for the readiness gap. AWS invests a fraction of its annual revenue in R&D and infrastructure—conservatively modelled at 10–12% of cloud segment revenue, implying annual capital deployment of roughly €9–11B. The combined annual investment of all five EU providers in this study is estimated to be below €2B, and even this figure is dominated by STACKIT’s announced multi-year programme, where annual deployment lags announced commitment by several years. The revenue analysis therefore establishes that the readiness gap is not a function of strategic misalignment; it is a function of economic scale that no individual EU provider can currently overcome through organic growth alone.

### 8.5.2 Readiness Gap Closure: Projected Parity Timelines

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*[Insert screenshot: Cloud Readiness Gap Closure Projection, 2006–2050]*

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Figure 8.3: Historical readiness trajectories (solid lines) modelled as linear growth from zero at each provider’s cloud launch year to the Q2 2026 average readiness score, with projections (dashed lines) at the same velocity to AWS parity (score = 100). Vertical markers indicate estimated parity years.

The readiness gap-closure model (Figure 8.3) applies the linear velocity assumption defined in Section 5.5.4 (Equations 5.5–5.6): each provider’s historical readiness improvement is modelled as uniform annual growth from zero at the year of cloud service launch to the average readiness score observed at Q2 2026, projected forward at the same rate to AWS parity (score = 100). The resulting parity years are structural benchmarks under a *ceteris paribus* assumption—stable investment levels, stable AWS readiness, and no disruptive capability jumps—and should not be interpreted as forecasts.

Under this model, T-Systems (OTC / T-Cloud Public), which combines the longest continuous operational history in the EU sovereign cloud cohort (OTC launched in 2016) with one of the higher readiness scores, is projected to reach parity with the AWS baseline at approximately 2029–2031. STACKIT, despite its lower 2026 readiness score reflecting its 2021 launch, achieves a comparable velocity owing to its substantial and accelerating capital programme; its parity projection falls in a similar range. These two providers represent the credible near-term contenders for AWS readiness parity under the existing investment trajectory.

OVHcloud, Scaleway, and IONOS present a more challenging picture. OVHcloud’s 2021 IPO improved its capital access but has not yet translated into the managed-service depth needed to close the AI/ML and database dimension gaps. Scaleway’s trajectory is constrained by its Iliad parent’s prioritisation of telecoms infrastructure capital. IONOS is hampered by its deliberate SME-segment positioning, which deprioritises the enterprise managed services that dominate the readiness scoring. For these three providers, parity timelines under the stable-velocity model range from the late 2030s to the mid-2040s—a timeframe that calls into question whether parity under the current capital regime is achievable before structural market consolidation reshapes the landscape.

The critical limitation of the linear model is that readiness improvement is unlikely to remain linear. Managed services exhibit platform economics: each new service lowers the cost of the next through shared infrastructure, shared APIs, and shared developer tooling. Providers that cross a critical managed-service density threshold may experience accelerating readiness improvement. STACKIT’s programme—which targets €11B of data centre infrastructure at Bitterfeld-Wolfen, with Phase 1 targeted for completion in 2027—represents precisely this kind of step-change investment that could non-linearly accelerate its readiness trajectory beyond what the velocity model predicts.

### 8.5.3 Investment Milestones and Capital Events

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*[Insert screenshot: Investment and Funding Milestone Timeline, 2012–2026]*

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Figure 8.4: Investment and funding milestones for AWS and the five EU providers, 2012–2026. Bubble area is proportional to the committed amount (EUR); colour encodes the provider. AWS milestones are included for scale reference.

The investment milestone timeline (Figure 8.4) contextualises the discrete capital events underlying the revenue and readiness trajectories. The most immediately visible feature is the scale of the STACKIT Bitterfeld-Wolfen data centre commitment (€11B, announced 2024)—the single largest capital pledge in the EU sovereign cloud cohort by a factor exceeding five. Without it, the EU cohort’s total disclosed capital across all providers and all years is in the range of €3–4B; with it, STACKIT accounts for the majority of total disclosed EU capital.

The OVHcloud 2021 IPO (€400M) and 2024 EIB green loan (€200M) are the next-largest EU events, reflecting a financing strategy that combines public market capital with European institutional debt. IONOS’s 2023 Frankfurt Stock Exchange listing raised approximately €480M—similarly structured. These market capital events occurred during a period of rising European interest in digital infrastructure, and both providers have used the resulting capital primarily for data centre expansion and managed-service development rather than acquisition.

The IPCEI-CIS-related milestones—while symbolically significant, as OVHcloud, Scaleway, and Deutsche Telekom are all official beneficiaries of the December 2023 Commission approval—are modest in absolute scale. Published national indicative funding envelopes for cloud IPCEI-CIS participants are in the range of €10–80M per beneficiary per member state, which is meaningful

for an early-stage capability investment but insufficient to materially close the revenue or readiness gap with AWS. The April 2026 EC Sovereign Cloud Tender (€180M across Lots 1–3) is in the same order of magnitude.

The AWS milestones included for comparison illustrate the asymmetry clearly: AWS’s capex commitments in Europe in 2023–2024 alone—announced infrastructure investments in Germany, Spain, and France—individually exceed the total lifetime disclosed capital of most EU providers in this study.

#### 8.5.4 Capital Efficiency: Readiness Points per Euro Invested

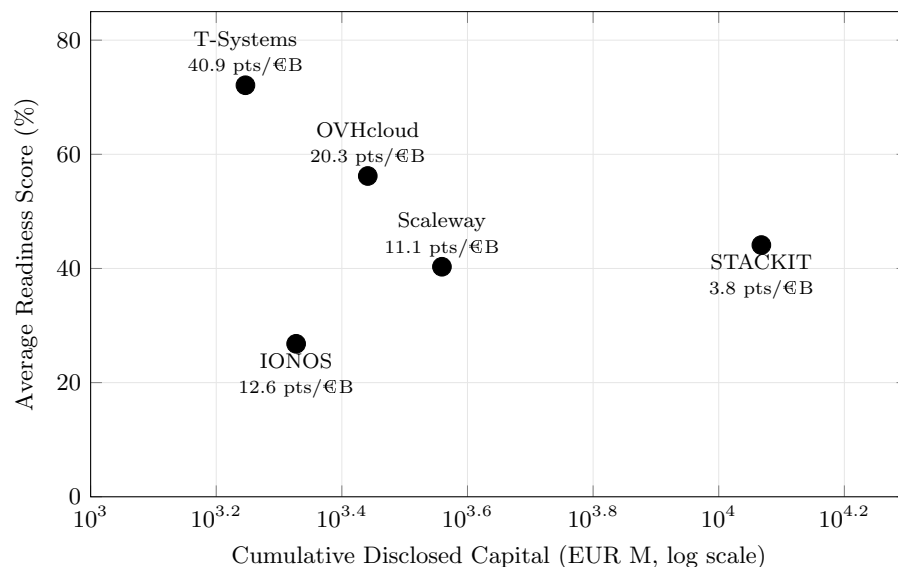


Figure 8.5: Investment efficiency scatter: each provider is plotted by cumulative disclosed capital (EUR M, log scale) on the X-axis and current average readiness score on the Y-axis. The efficiency ratio (readiness points per €B of disclosed capital) is shown beneath each label. STACKIT’s low ratio reflects that its €11B Bitterfeld-Wolfen commitment is a forward pledge, not yet fully deployed.

The capital efficiency scatter (Figure 8.5) relates cumulative disclosed capital to current average readiness score for each EU provider, surfacing the degree to which capital has translated into measurable capability. The efficiency ratio  $\eta$  is defined in Section 5.5.4 (Equation 5.7); only capital events with a publicly confirmed nominal amount contribute to the denominator, with undisclosed events retained in the milestone timeline but excluded from all quantitative aggregations. The ratio reveals a pattern that is counterintuitive given the naive expectation that higher investment produces higher readiness.

T-Systems (OTC / T-Cloud Public) and IONOS emerge as the most capital-efficient providers in the cohort. T-Systems achieves its high readiness score through a combination of the mature OTC managed-service catalogue (developed since 2016 with continuous parent investment from Deutsche Telekom) and the purpose-built sovereign architecture of T-Cloud Public (purpose-built

Biere and Magdeburg infrastructure). IONOS achieves comparable efficiency through decades of organic growth in its managed hosting and cloud business, reflecting operational experience that compounds without requiring step-change capital events.

STACKIT presents the apparent paradox of high cumulative disclosed capital and a *lower* current readiness score than T-Systems or IONOS. The explanation is structural: STACKIT’s disclosed capital is dominated by the €11B Bitterfeld-Wolfen commitment, which is a forward-looking infrastructure pledge, not historical operational expenditure. The capital has been committed but not yet deployed in full; Phase 1 construction is ongoing as of Q2 2026 and the associated managed-service layer has not yet been built on top of the physical infrastructure. Once the infrastructure is commissioned and the managed-service stack matures—a timeline STACKIT’s own roadmap places in the 2027–2029 period—its efficiency ratio is expected to improve substantially.

OVHcloud and Scaleway occupy intermediate positions: moderate disclosed capital, moderate readiness scores, and efficiency ratios that reflect their role as full-featured providers operating without either STACKIT’s capital concentration or T-Systems’ legacy managed-service depth.

### 8.5.5 Capital Structure and its Sovereignty Implications

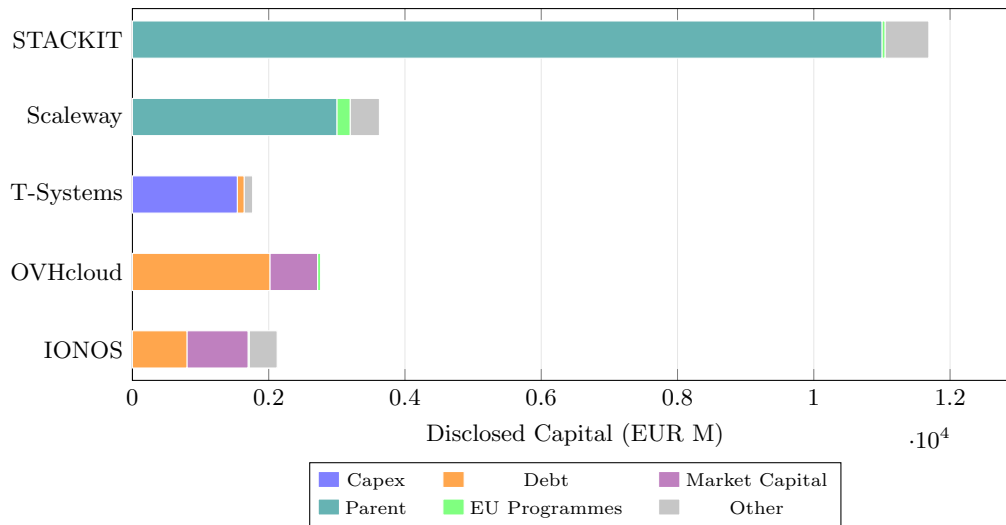


Figure 8.6: Capital structure decomposition for the five EU providers, showing cumulative disclosed capital by funding source category (EUR M). T-Systems abbreviates T-Systems (OTC / T-Cloud Public). STACKIT’s bar is dominated by the €11B Schwarz Group parent pledge at Bitterfeld-Wolfen. OVHcloud and IONOS are the only two providers with US-managed-fund exposure in the Market Capital and Debt categories.

The capital structure decomposition (Figure 8.6) reveals qualitatively different financing strategies across the EU provider cohort—strategies that are not neutral with respect to sovereignty scoring under the SovScore framework.

STACKIT’s capital stack is dominated by Parent funding: the Schwarz Group’s discretionary investment commitments constitute nearly all of STACKIT’s disclosed capital base. The Schwarz Group

is a German family-owned group with no listed equity and no US institutional investor exposure. This capital structure is sovereignly ideal: it creates no governance vectors through US law, no CLOUD Act exposure through US parent ownership, and no shareholder pressure structures that could be leveraged through US jurisdiction. SOV-1 Q3 (“Is EU-based financing the primary source?”), SOV-2 Q1 (“No compelled access under CLOUD Act?”), and SOV-2 Q2 (“No US PE/VC investor with board-level influence?”) all score Y for STACKIT.

Scaleway’s capital structure is similarly dominated by Parent funding from Iliad Group—a French-listed telecoms group with no material US shareholder or debt exposure. Like STACKIT, Scaleway scores Y on SOV-2 Q2, as no US PE/VC investor holds board-level influence.

T-Systems (OTC / T-Cloud Public) is funded through Capex transfers from Deutsche Telekom AG, a DAX-listed entity with international institutional shareholders but with primary governance under German law and no US PE control. Its capital structure triggers no CLOUD Act exposure, and its SOV-2 scores reflect this.

OVHcloud and IONOS present a more complex picture. OVHcloud’s Market Capital category includes a combined €500M private equity injection from KKR and General Atlantic in 2021–2022—US-domiciled funds with potential board-level influence and US legal exposure. IONOS’s Market Capital includes the 2023 Frankfurt IPO and earlier PE investment from Warburg Pincus (US-domiciled). Both of these capital structures create governance vectors that the SovScore methodology explicitly penalises: under SOV-2 Q2, any US PE/VC investor with board-level influence produces a P (Partial) answer, reflecting that although cloud operations are EU-hosted, the investor governance chain touches US legal jurisdiction. Neither OVHcloud nor IONOS achieves a Y on SOV-2 Q2, and this outcome is a direct function of their capital structure.

### 8.5.6 Capital Source as a Direct Sovereignty Determinant

The preceding analysis establishes that the relationship between capital structure and sovereignty score in the SovScore framework is not correlational but causal. The methodology, as designed in Section ??, contains questions that directly score the legal and governance implications of capital origin:

- **SOV-1 Q3** (“Is financing primarily from EU institutions?”) awards Y only where the primary capital source is EU-domiciled, producing a direct positive contribution to the SEAL 1 (Data Sovereignty) seal score. STACKIT, Scaleway, and T-Systems score Y; OVHcloud and IONOS score P owing to their US PE/market capital components.
- **SOV-2 Q1** (“Is the provider free from CLOUD Act compelled disclosure?”) awards Y where no group entity is subject to US jurisdiction under 18 U.S.C. §2523—the Electronic Communications Privacy Act amendment commonly called the CLOUD Act. IONOS’s US subsidiary IONOS Inc. (registered in Pennsylvania) creates a potential exposure vector, resulting in a P for IONOS. Providers without any US-registered entity in the corporate chain score Y.
- **SOV-2 Q2** (“Does the shareholder structure contain no US PE/VC investor with board-level influence?”) directly scores capital structure as a governance vector. KKR’s and Warburg Pincus’s involvement in OVHcloud and IONOS respectively produce P answers for both, reducing their SEAL 2 (Legal Sovereignty) seal scores.
- **SOV-2 Q3** (“Is the provider free from any parent-company jurisdiction outside the EU?”) intersects capital structure in cases where a parent-level entity that is itself a capital source

is subject to non-EU law. OVHcloud’s 2022 Ontario Superior Court ruling—which ordered OVHcloud to comply with a data-disclosure order—produced a N on this question, the most penalising outcome in the SEAL 2 seal, and is directly traceable to the jurisdictional exposure created by OVHcloud’s pre-IPO international expansion financed through private equity.

This scoring structure means that capital from US private equity—regardless of the cloud service’s technical architecture or data residency—directly reduces a provider’s SovScore through governance and jurisdiction vectors. Conversely, parent-group capital from EU-domiciled, non-US-exposed entities (as with STACKIT and Scaleway) produces the most favourable sovereignty scores, all else being equal. The financial analysis therefore arrives at a finding with direct policy implications: *EU sovereign cloud investment is most effective, from a sovereignty standpoint, when it originates from EU-controlled entities*—public institutions (IPCEI-CIS, EIB, EC tender awards), EU-listed parent groups, or EU family-owned conglomerates—rather than international capital markets where US jurisdictional exposure is difficult to quarantine from the operating entity. The modest scale of IPCEI-CIS and the EC tender relative to the investment needed to close the readiness gap with AWS indicates that this alignment remains more aspiration than structural reality for the full EU cloud cohort.

## 8.6 Performance vs. Sovereignty: Structural Constraints

Two SOV verdicts are not provider-specific failures: they are industry-wide structural constraints that affect every provider in the study and that cannot be resolved through governance decisions alone.

### 8.6.1 SOV-6 Q4: No EU HPC Silicon

SOV-6 Q4 assesses whether the provider’s high-performance computing infrastructure uses silicon from EU-governed or trusted sources. Every provider in the study receives a negative verdict on this question. The reason is straightforward: there is no commercially deployed EU-origin general-purpose GPU or high-performance CPU that is competitive with Intel Xeon, AMD EPYC, or NVIDIA data-centre GPUs. The European Processor Initiative (EPI) has produced the RHEA chip and is developing SiPearl processors for HPC, but these are not yet available as commercial cloud infrastructure components. Arm-based EU-assembled chips exist but do not constitute an EU-IP supply chain at the silicon level.

This structural constraint sets a practical ceiling on SOV-6 scores for all providers irrespective of their other governance choices. It is a limitation of the current assessment that it assigns a negative verdict to all providers equally without distinguishing between those actively diversifying their HW procurement (T-Systems’ dual-vendor strategy) and those with no such plans. Future iterations of the framework might introduce a partial verdict for providers participating in EU HPC silicon transition programmes.

### 8.6.2 SOV-5 Q2: Non-EU Firmware

SOV-5 Q2 assesses whether server firmware is of EU origin. Again, every provider receives a negative verdict because Intel Management Engine firmware, AMD Platform Security Processor firmware, and NVIDIA GPU firmware are all developed and controlled by US-headquartered corporations.

These firmware components run below the operating system and have privileged access to system hardware.

This is not a theoretical risk. The security research community has documented multiple instances where firmware-level access vectors have been exploited [16]. The EU CHIPS Act and the Open Compute Project’s Caliptra initiative represent long-term policy responses to this dependency, but neither produces EU-origin firmware on a horizon relevant to current cloud procurement decisions.

For the purposes of interpreting SovScore results, assessors should treat SOV-5 Q2 and SOV-6 Q4 as systemic floor constraints rather than provider-discriminating findings. Providers that score well on all other SOV-5 and SOV-6 questions despite these universal negatives are effectively doing as well as any EU provider can do under current industry conditions.

### 8.6.3 Policy Instruments

Three EU policy instruments are relevant to the long-term resolution of these structural constraints. The **EU CHIPS Act** (2023) targets €43 billion in semiconductor investment to bring leading-edge chip manufacturing to European soil by 2030. The **European Processor Initiative** is developing the SiPearl HPC processor based on Arm architecture for the EuroHPC supercomputer programme. The **EU RISC-V initiative** supports the development of open-ISA processors that could eventually underpin EU-origin cloud server silicon free of US export-control risk. None of these initiatives will produce commercial cloud-grade silicon within the thesis horizon, but they establish the infrastructure for a future re-assessment in which SOV-6 Q4 might be satisfiable.

## 8.7 Vendor Lock-in Implications

The Cloud Readiness Assessment was designed around a specific set of 60 services corresponding to Alps Alpine’s enterprise workload requirements. A secondary reading of the gap data reveals a vendor lock-in dynamic that has implications both for migration strategy and for the interpretation of sovereignty.

### 8.7.1 OpenStack as an Interoperability Anchor

Providers that build their IaaS layer on OpenStack — including T-Systems OTC, Cleura, Elastx, and Fuga Cloud — expose standard OpenStack APIs for compute (Nova), storage (Swift, Cinder), networking (Neutron), and identity (Keystone). This means that workloads containerised in Kubernetes and deployed against standard OpenStack APIs are, in principle, portable between these providers and between cloud and on-premise environments. The GAIA-X federated architecture framework [18] explicitly targets this interoperability as a design goal.

The irony is that OpenStack-based providers tend to have lower readiness scores in dimensions that require proprietary managed services (AI/ML, serverless, managed analytics), precisely because they have not built the proprietary PaaS layers that create lock-in. A provider that offers only OpenStack IaaS is maximally portable and minimally complete relative to the 60-service baseline.

### 8.7.2 Proprietary Managed Services and Lock-in

The highest-readiness EU providers have built proprietary managed services on top of their IaaS foundations. T-Systems OTC's ModelArts (managed ML), GaussDB (managed NewSQL), and DCS (managed Redis) are services with no simple migration path to another provider. OVHcloud's managed database portfolio, similarly, creates dependency once workloads are running against provider-specific APIs or data formats.

This creates a layered lock-in structure: at the IaaS layer, OpenStack portability is high; at the managed PaaS layer, provider-specific APIs create stickiness; at the AI/ML layer, model management, training pipelines, and inference endpoints are deeply provider-specific.

### 8.7.3 The Migration Barrier from AWS

The gap table data makes explicit what cloud migration practitioners know empirically: switching from AWS to any EU provider requires architectural re-engineering of any workflow that depends on AWS-proprietary services. DynamoDB's serverless scaling model has no direct analogue in EU providers; Step Functions' workflow orchestration would need to be replaced by open-source alternatives (Apache Airflow, Temporal) that are not offered as managed services by most EU providers; Bedrock's model API has no EU substitute. For an enterprise currently running on AWS, the effective migration cost is dominated not by the commodity services (compute, storage, networking) but by these proprietary integrations.

This observation supports the recommendation (Section 8.8) for a hybrid strategy in which sovereignty-sensitive and commodity workloads migrate to EU providers while AI/ML and specialised managed-service workloads remain on AWS until EU equivalents mature.

## 8.8 Recommendations for EU Enterprises

The following four strategic postures emerge from the combined findings of the SovScore and Cloud Readiness Assessment. They are not mutually exclusive; most enterprises will benefit from a combination of approaches.

### 8.8.1 Strategy 1: Lift-and-Shift to a High-Readiness EU Provider

For enterprises in regulated industries — financial services, healthcare, public administration — operating workloads that currently run on AWS IaaS and standard managed databases (PostgreSQL, MySQL, Kubernetes), migration to T-Systems OTC or OVHcloud represents the path of least resistance. Both providers offer coverage of core IaaS and PaaS services sufficient to support the majority of enterprise workloads without architectural re-engineering. Both hold BSI C5 or equivalent certifications, GDPR data-processing agreements, and EU data residency guarantees. The readiness gap for these workloads is manageable.

The sovereign risk trade-off for T-Systems OTC (Huawei dependencies) should be assessed against the enterprise's specific threat model. For German public administration workloads, the GovTech approval of T Cloud Public provides regulatory cover. For enterprises with higher sensitivity to China-related extraterritorial risk, OVHcloud or IONOS may be preferable.

### 8.8.2 Strategy 2: Sovereignty-First Selection for Sensitive Workloads

For workloads handling sensitive personal data, legally privileged information, or national-security-adjacent content, legal certainty outweighs managed-service richness. Infomaniak (Swiss law, FADP compliance, no non-EU parent), Elastx (Swedish law, OpenStack, ISO 27001), and Cleura (Swedish law, OpenStack, dedicated sovereignty tier) offer the cleanest sovereignty profiles in the study. The trade-off is an acceptance of a thinner managed-service portfolio and the operational overhead of self-managing services that AWS or OTC would deliver as fully managed.

This posture is most appropriate for legal firms, medical data processors, and public authorities handling data categories under GDPR Article 9, where the consequences of a Schrems II-type legal challenge to the processing arrangement outweigh the operational cost of a more limited cloud platform [6].

### 8.8.3 Strategy 3: Hybrid Cloud Architecture

The most pragmatic path for the majority of EU enterprises is a deliberate hybrid architecture that routes workloads by sovereignty sensitivity. Sovereignty-sensitive workloads — employee data, customer PII, regulated financial records — run on a high-sovereignty EU provider. Commodity development infrastructure and AI/ML workloads run on AWS (or, where appropriate, on Azure or GCP), accepting the sovereignty trade-off in return for access to services with no current EU equivalent.

This architecture requires careful data-flow boundary design to ensure that personally identifiable data does not transit from the EU-provider environment to the AWS environment. It is technically achievable with standard networking and identity federation tooling and is consistent with the ENISA guidance on hybrid cloud sovereignty [16].

### 8.8.4 Strategy 4: Monitor STACKIT for 2027–2028 Re-evaluation

STACKIT currently has the lowest scalability score (32) of the providers seriously considered for enterprise adoption. However, its trajectory is uniquely well-resourced: Schwarz Group’s financial scale, its captive enterprise demand, and its €300M+ investment programme make it the EU provider most likely to show dramatic readiness improvement over the 2026–2030 horizon. Enterprises making cloud-infrastructure decisions in 2026 should include STACKIT in their evaluation criteria for 2028 contract renewals or new workload deployments. If STACKIT’s managed-service catalogue matures as its investment suggests, it would combine high readiness with a straightforward EU corporate structure free of the Huawei complications that affect T-Systems.

## 8.9 Limitations of This Study

### 8.9.1 Self-Reported and Publicly Documented Evidence Base

Both the SovScore assessments and the Cloud Readiness verdicts are based on publicly available documentation, provider websites, compliance certificates, and — for selected providers — responses to direct enquiry. No independent technical audit was conducted. A provider that falsely represents its service capability or its supply-chain composition in public documentation would receive an unwarranted favourable verdict. This limitation is inherent to any study of this scope covering 18

providers across 8 sovereignty objectives and 60 services; independent auditing at this scale would require regulatory authority and resource access far beyond an academic research project.

### 8.9.2 Static Snapshot in a Rapidly Evolving Market

Provider service catalogues, corporate structures, and compliance certifications change continuously. All verdicts in this study reflect the state of each provider as documented in Q1–Q2 2026. Several providers have announced managed services or corporate-structure changes that postdate the assessment window. T-Systems’ T Cloud Public migration is the most significant of these; the analysis in Section 8.3 treats it as a prospective development rather than a completed transition because the OTC-to-TCP migration was not complete at the assessment date.

### 8.9.3 AWS Baseline Bias

Treating AWS as the 100% readiness baseline implicitly privileges the hyperscaler managed-service model as the normative reference point for enterprise cloud capability. EU providers operating on OpenStack IaaS may offer equivalent functional capability through different architectural patterns — for example, a self-managed Apache Kafka deployment may satisfy the same enterprise requirement as Amazon MSK — but would receive a partial or negative verdict in the readiness assessment because the service form differs from the AWS reference.

This bias is a deliberate design choice: the assessment is calibrated to the services that Alps Alpine’s enterprise workload requires in the form it currently uses them. It reflects a real-world migration scenario rather than an abstract capability comparison. But it should not be read as a claim that OpenStack IaaS is intrinsically inferior; it means that migrating an existing AWS-pattern workload to an OpenStack provider requires architectural adaptation that entails cost and risk [3].

### 8.9.4 Service Selection Tailored to a Specific Enterprise

The 60-service taxonomy was constructed from Alps Alpine’s cloud workload requirements across thirteen dimensions. A different enterprise with different workloads — a media company, a financial institution, a public administration body — would produce a different service selection and potentially a different ranking of providers. Providers that score poorly on the scalability and AI/ML dimensions relevant to Alps Alpine might score well on dimensions more relevant to other industries (e.g., managed SAP environments, HPC for simulation workloads). The results should be interpreted as specific to the Alps Alpine enterprise context rather than as a universal EU-cloud ranking.

### 8.9.5 No SLA or Operational Quality Dimension

Coverage verdicts capture the existence of a service, not its operational maturity, reliability, latency profile, or supported SLA. A managed Kubernetes service offered by a small EU provider with a 99.5% SLA and a 24-hour support window is counted the same as AWS EKS with a 99.95% SLA and a 15-minute critical-incident response. The readiness scores therefore overstate the practical substitutability of EU services for enterprises with high availability requirements. A future extension of this study could introduce a quality dimension — drawing on published SLA data, Uptime Institute reports, and third-party latency benchmarks — to complement the coverage assessment [2].

### 8.9.6 Exclusion of Cost and Commercial Terms

Cloud infrastructure cost is a significant procurement factor that this study does not address. EU providers frequently price their services at a premium to AWS for equivalent compute and storage, reflecting their smaller economies of scale. This premium may be acceptable or even welcome in procurement frameworks where EU origin is a requirement (IPCEI, GovTech), but it is a material constraint for cost-sensitive workloads. A cost-adjusted readiness assessment — weighting service availability by the cost premium relative to AWS — would produce a more complete picture of the EU cloud procurement decision.

## Chapter 9

# Conclusion

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### 9.1 Summary of Findings

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### 9.2 Contribution to Research

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### 9.3 Future Work

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## Appendix A

# Alps Alpine Infrastructure Checklist (Full)

## Appendix B

# Europeanness Scoring Matrix

## Appendix C

# Benchmark Raw Data

## Appendix D

# Terraform Provisioning Scripts

## Appendix E

# Operational Friction Log (Full)

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